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UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

MURRAY & POOLE ENTERPRISES LTD.,
Petitioner,

v.

INSTITUT DE CARDIOLOGIE DE MONTREAL,
Patent Owner.

IPR2023-01064
Patent 11,400,063 B2

Before ULRIKE W. JENKS, SHERIDAN K. SNEDDEN, and ZHENYU YANG, *Administrative Patent Judges*.

JENKS, *Administrative Patent Judge*.

JUDGMENT
Final Written Decision Determining
All Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Murray & Poole Enterprises Ltd. ("Petitioner") filed a Petition requesting an *inter partes* review of claims 1–50 ("the challenged claims") of U.S. Patent No. 11,400,063 B2 (Ex. 1001, "the '063 patent"). Paper 1 ("Petition" or "Pet."). We instituted trial on January 16, 2024. Paper 9 ("Decision" or "Dec."). Institut de Cardiologie de Montréal ("Patent Owner") filed a Patent Owner Response to the Petition. Paper 21 ("PO Resp."). Petitioner filed a Reply (Paper 34 ("Reply")) and Patent Owner filed a Sur-reply (Paper 37 ("Sur-reply")). An oral hearing was held on October 15, 2024, and a transcript is of record. Paper 48 ("Tr.").

We have jurisdiction under 35 U.S.C. § 6(b). After considering the full record developed through trial, we determine that Petitioner has proved by a preponderance of the evidence that the challenged claims are unpatentable. *See* 35 U.S.C. § 316(e). Our reasoning is explained below, and we issue this Final Written Decision under 35 U.S.C. § 318(a).

A. Real Parties in Interest

Petitioner identifies Murray & Poole Enterprises Ltd., AGEPHA Pharma FZ LLC, AGEPHA Pharma USA, LLC, and UPTTON Limited, as real parties-in-interest. Pet. 72. Patent Owner identifies Institut de Cardiologie de Montréal ("MHI") and Pharmascience Inc. as real parties-in-interest. Paper 5, 2.

B. Related Matters

The parties state that they are not aware of any related matters. Pet. 72; Paper 5, 2.

C. The ’063 Patent (Ex. 1001)

The ’063 patent is titled “Early Administration of Low-dose Colchicine After Myocardial Infarctions.” Ex. 1001, code (54). The ’063 patent issued from Application No. 17/097,285 (“the ’285 application”), filed November 13, 2020. *Id.* at codes (21), (22). The ’063 patent claims priority to Provisional Application No. 63/093,988 (“the ’988 provisional application”), filed on October 20, 2020 and Provisional Application No. 62/935,865 (“the ’865 provisional application”), filed on November 15, 2019. *Id.* at code (60). The ’063 patent relates to “a method of treating a patient after having a myocardial infarction (MI), the method including initiating the administration of colchicine at a daily low dose to the patient within about 3 days of the MI.” *Id.* at code (57).

According to the ’063 patent, “[i]nflammation appears to play an important role in atherosclerosis . . . and coronary artery disease.” *Id.* at 1:9–11. The ’063 patent states that “[b]ecause acute coronary syndromes are associated with higher risks of recurrent events and exacerbated inflammation a need exists in the art for new treatment regimens.” *Id.* at 1:60–62. The ’063 patent explains that “[c]olchicine is an inexpensive, orally administered, potent anti-inflammatory medication” that is indicated for the treatment of gout, familial Mediterranean fever, and pericarditis. *Id.* at 1:28–30, 1:43–47. However, the cardiovascular effects of colchicine have also been evaluated in trials in which participants were administered 0.5 mg of colchicine once daily. *Id.* at 1:52–2:28. In one such example, trial participants with stable coronary disease were treated with colchicine at a dose of 0.5 mg once daily and had fewer cardiovascular events than those not receiving colchicine. *Id.*

In a second example, patients who had recently suffered an MI were treated with colchicine at a dose of 0.5 mg once daily and had a statistically significant lower risk of ischemic cardiovascular events than placebo; had significantly reduced rates of death from cardiovascular causes, resuscitated cardiac arrest, myocardial infarction, stroke, and urgent hospitalization for angina leading to coronary revascularization compared to placebo; and had reduced morbidity relative to placebo. *Id.* In that study, colchicine reduced the risk of ischemic cardiovascular ("CV") events by 23% in the post-MI setting. *Id.* at 2:41–43.

In a Table 3, the time to treatment initiation ("TTI") was evaluated and patients who had recently suffered an MI were treated with colchicine at a dose of 0.5 mg once daily starting on days post-MI from 0–3, 4–7, and 8–30. *Id.* at 12:34–68. In that TTI study, initiation of colchicine treatment in days from 0–3 post-MI reduced the risk of the composite primary endpoint by 48%. *Id.* at 2:43–47. The composite primary endpoint consisted of "a composite of CV death, resuscitated cardiac arrest, MI, stroke, or urgent hospitalization for angina requiring coronary revascularization." *Id.* at 8:5–8. The invention claimed in the '063 patent is based on the treatments used in the TTI study.

1. *Overview of Provisional Patent Application No. 63/093,988 (Ex. 1003)*

The '988 provisional application describes patients with coronary artery disease (CAD) as being at "an increased risk of cardiovascular events including death, myocardial infarction (MI), stroke, and cardiac arrest; a risk that is magnified for each subsequent cardiovascular event." Ex. 1003, 2:3–4 (footnote omitted). "[I]nflammation has emerged as an important therapeutic target in patients with atherosclerosis." *Id.* at 2:12–13 (footnote omitted). The '988 provisional application is directed to a contemporary meta-analysis evaluating randomized control patient trials comparing colchicine to placebo (or no colchicine) for secondary cardiovascular prevention. *Id.* at 7:21–22. "Colchicine has an established safety profile from its centuries of use to treat gout." *Id.* at 10:5.

The colchicine trials show improvement in patients with CAD.

> In patients with CAD, the addition of colchicine to standard medical therapy was associated with a statistically significant reduction in the primary composite endpoint of cardiovascular mortality, MI, ischemic stroke, and urgent coronary revascularization, compared to patients on placebo or no colchicine.

Ex. 1003, 6:8–12. "[A]ll trials showed a decrease in urgent coronary revascularizations among patients randomized to colchicine, [but] only COLCOT[1] detected a reduction in ischemic strokes." *Id.* 8:11–13.

The '988 provisional application explains that

> [t]he anti-inflammatory properties of colchicine work through various mechanisms to inhibit the pathogenesis of CAD, and subsequently reduce the incidence of ischemic cardiovascular events. In addition to targeting the NLRP3 inflammasome leading to the reduction of circulating levels of IL-1β and IL-6, colchicine also inhibits cholesterol crystals that promote inflammation and plaque instability in atherosclerosis.

Ex. 1003, 9:3–7 (footnote omitted).

[1] Colchicine Cardiovascular Outcomes Trial (COLCOT) (Ex. 1006).

2. Overview of Provisional Patent Application No.62/935,865 (Ex. 1004)

The ’865 provisional application describes colchicine as “an inexpensive, orally administered, potent anti-inflammatory medication that was initially extracted from the autumn crocus and has been used for centuries.” Ex. 1004, 1:21–22.

The ’865 provisional application explains that “[i]n the Low-Dose Colchicine (LoDoCo) trial, patients with stable coronary disease treated with colchicine at a dose of 0.5 mg once daily had fewer cardiovascular events than those not receiving colchicine.” *Id.* at 1:36–38. In a different clinical trial referred to as the COLCOT “the effects of colchicine on cardiovascular outcomes as well as its long-term safety profile in patients who had recently had a myocardial infarction” was evaluated. *Id.* at 2:8–10.

The ’865 provisional application describes that “the method involves administering colchicine within 5, 10, 15, 20, or 25 days of the MI.” *Id.* at 2:32–33; *see also id.* at 9 (Table 1 – Patients were enrolled a mean 13.4±10.2 days after myocardial infarction). The ’865 provisional application discloses that “colchicine is administered at 0.4 to 0.6 mg, and is preferably administered at 0.5 mg” daily. *Id.* at 3:12–13.

The ’865 provisional application concludes that colchicine administered at a daily low dose of 0.5 mg led to a statistically significant lower risk of ischemic cardiovascular events than placebo. Death from cardiovascular causes, resuscitated cardiac arrest, myocardial infarction, stroke, or urgent hospitalization for angina leading to coronary revascularization was also significantly lower among the patients who received 0.5 mg of colchicine than those who received placebo. Ex. 1004, 2:13–17.

D. Illustrative Claims

Claims 1, 8, and 27 are the independent claims challenged by Petitioner in this proceeding. Independent claims 1, 8, and 27, reproduced below, are illustrative of the subject matter:

> 1. A method of treating a patient after having a myocardial infarction (MI), [1a] the method comprising initiating the administration of colchicine at a daily low dose to the patient [1b] within about 3 days of the MI, [1c] wherein the colchicine is administered without pre-loading the patient with a higher dose of colchicine before the administration at the daily low dose, [1d] wherein percutaneous coronary intervention was performed for treating the patient's MI.

Ex. 1001, 41:29–36 (numbering and bracketing added for reference convenience).

> 8. A method of treating a patient a[f]ter having a myocardial infarction (MI), [8a] the method comprising initiating the administration of colchicine at a daily low dose to the patient [8b] within about 3 days of the MI, [8c] wherein the colchicine is administered without pre-loading the patient with a higher dose of colchicine before the administration at the daily low dose, [8d] wherein the patient is at a lower risk of a cardiovascular event, relative to a patient not being administered colchicine, and [8e] wherein the cardiovascular event is resuscitated cardiac arrest, stroke, or urgent hospitalization for angina requiring coronary revascularization.

Ex. 1001, 41:50–60 (numbering and bracketing added for reference convenience).

> 27. A method of reducing the risk of a stroke in a patient after having an MI, the method comprising initiating the

administration of colchicine at a daily low dose to the patient within about 30 days of an MI.

Ex. 1001, 42:33–36.

E. Asserted Grounds

Petitioner asserts that claims 1–50 of the ’063 patent are unpatentable based on the following grounds. Pet. 24.

Would understand the preamble Mutation Claims challenged	**35 U.S.C. §**	**References/Basis**
27, 29–36, 39	102	COLCOT protocol[2]
1–50	103	Nidorf,[3] COLCOT Protocol, ESC Guidelines,[4] COLCRYS[5]
1–13, 17–26, 28, 40–46, 49, 50	102	Bouabdallaoui[6]
1–26, 28, 40–50	103	Bouabdallaoui, COLCRYS

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[2] "*Colchicine Cardiovascular Outcomes Trial (COLCOT) (COLCOT)*," https://web.archive.org/web/20171020004013/https:/clinicaltrials.gov/ct2/show/NCT02551094. Ex. 1006 ("COLCOT protocol").

[3] Nidorf, US 10,206,891 B2, issued Feb. 19, 2019. Ex. 1007 ("Nidorf").

[4] Ibanez et al., *2017 ESC Guidelines for the management of acute myocardial infarction in patients presenting with ST-segment elevation*, EUROPEAN HEART J., 39:119–177 (2018). Ex. 1009 ("ESC guidelines").

[5] Package insert for COLCRYS™ (colchicine, USP) tablets for Oral use, *available at* https://www.accessdata.fda.gov/scripts/cder/daf/index.cfm?event=overview.process&App1No=022351. Ex. 1016 ("COLCRYS").

[6] Bouabdallaoui et al., *Time-to-treatment initiation of colchicine and cardiovascular outcomes after myocardial infarction in the Colchicine Cardiovascular Outcomes Trial (COLCOT)*, EUROPEAN HEART J., 41:4092–4099 (2020). Ex. 1005 ("Bouabdallaoui").

Petitioner submits testimony from Paul Ridker, M.D. Ex. 1049 (Ridker Decl.); Ex. 1075 (Supplemental Ridker Decl.). In addition, Petitioner also submits testimony from Raymond Cruitt M.S. and James L. Mullins, Ph.D. discussing the public availability of certain references. Ex. 1038 (Cruitt Decl.); Ex. 1072 (Second Cruitt Decl.); Ex. 1073 (Mullins Decl.). Patent Owner responds with testimony from Hossein Ardehali M.D. Ex. 2001 (Ardehali Decl.); Ex. 2072 (Supplemental Ardehali Decl.). The deposition testimony of Drs. Ridker and Ardehali is also of record. Ex. 2071 (Ridker transcript); Ex. 1076 (Ardehali transcript).

In addition to the testimony from the parties' designated experts above, the record also includes testimony from two fact witnesses: Jean-Claude Tardif, M.D. (Ex. 2011 (Tardif Decl.)), (Ex. 1077 (Tardif transcript)); Nadia Bouabdallaoui, M.D., Ph.D. (Ex. 2068 (Bouabdallaoui Decl.)), (Ex. 1078 (Bouabdallaoui transcript)).

II. ANALYSIS

A. Claim Construction

The Board applies the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b). 37 C.F.R. § 100(b). Under that standard, claim terms "are generally given their ordinary and customary meaning" as understood by a person of ordinary skill in the art at the time of the invention. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc) (quoting *Vitronics Corp. v. Conceptronic, Inc.*, 90 F.3d 1576, 1582 (Fed. Cir. 1996)). "In determining the meaning of the disputed claim limitation, we look principally to the intrinsic evidence of record, examining the claim language itself, the written description, and the prosecution history, if in evidence." *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 469 F.3d 1005, 1014 (Fed. Cir. 2006) (citing *Phillips*, 415 F.3d at 1312–17).

Petitioner proposes constructions for several terms, including the preamble of claims 1–26 and 40–50 which recites "a method of treating a patient after having a myocardial infarction (MI)," the preamble of claims 27–39 which recites "reducing the risk of a stroke," and "wherein the patient is at a lower risk of a cardiovascular event" recited in claims 8–11 and 20–23. *See* Pet. 21–24.

Patent Owner responds to Petitioner's constructions of "reducing the risk of a stroke" and "wherein the patient is at a lower risk of a cardiovascular event," but does not discuss Petitioner's construction of "a method of treating a patient after having a myocardial infarction (MI)." PO Resp. 6–13.

1. *"a method of treating a patient after having a myocardial infarction (MI)"*

Petitioner contends that the preamble limitation is directed to treating a patient population but argues that there is no required clinical efficacy. Pet. 22 (citing *In re Montgomery*, 677 F.3d 1375, 1380 (Fed. Cir. 2012) ("We are skeptical that a proper interpretation of [the claim term 'for the treatment or prevention of stroke or its recurrence'] would include an efficacy requirement"); Ex. 1048 ¶¶ 97–104).

Patent Owner does not address this limitation. *See generally* PO. Resp.

We agree with Petitioner that the preamble does not have an efficacy requirement. Accordingly, we adopt Petitioner's interpretation that the preamble simply describes the patient population that is to receive treatment.

2. "reducing the risk of a stroke"

Petitioner contends that the limitation of "reducing the risk of a stroke" is an intended result. Pet. 23 (citing *Bristol-Myers Squibb Co. v. Ben Venue Lab'ys, Inc.*, 246 F.3d 1368, 1375–76 (Fed. Cir. 2001) (finding "a method for treating a cancer patient to effect regression of a taxol-sensitive tumor, said method being associated with reduced hematologic toxicity" "is only a statement of purpose and intended result"); Ex. 1049 ¶¶ 104–108).

Patent Owner contends that "the preamble [of reducing the risk of stroke in claim 27] is limiting by demonstrating that the reduction of risk of certain cardiovascular events, including the reduction of stroke, in patients who have suffered an MI is the 'essence of the invention.'" PO Resp. 7 (citing *Boehringer Ingelheim Vetmedica, Inc. v. Schering-Plough Corp.*, 320 F.3d 1339, 1345 (Fed. Cir. 2003)). In addition, Patent Owner contends that in the notice of allowability, "Examiner explicitly stated that a reason for allowance was that the prior art did not teach, show or suggest or make obvious the reduction in the risk of stroke." PO Resp. 8 (citing Ex. 1002, 399). Patent Owner argues that reliance on the preamble during prosecution transforms the preamble limitation into a claim limitation. PO Resp. 8 (citing *Invitrogen Corp. v. Biocrest Mfg., L.P.*, 327 F.3d 1364, 1370 (Fed. Cir. 2003)). Patent Owner contends that "Examiner correctly determined that the preamble is limiting because the claimed method without the preamble is meaningless." *Id.* at 10.

We are not persuaded by Patent Owner's contention. The facts in this case differ from the facts in *Invitrogen*. In *Invitrogen*, the applicant amended the preamble to distinguish the prior art, thus clearly indicating that Examiner was relying on the preamble to distinguish the claim over the prior art. *Invitrogen Corp. v. Biocrest Mfg., L.P.*, 327 F.3d at 1370. The preamble of claim 27 recites "[a] method of reducing the risk of a stroke in a patient after having an MI." Ex. 1001, 42:33–34. The body of the claim, however, does not recite that colchicine is administered to only those patients at risk of stroke. Ex. 1001, 42:34–36. Indeed, there is no indication in the specification that patients were selected for the trial based on their risk of stroke.[7] *See* Ex. 1001, 27:30–58, 32:19–67. At no point during prosecution the '063 patent did Examiner apply art against claim 27[8] necessitating any claim amendments by Patent Owner that would indicate Examiner relied on the preamble language to allow the claim. *See* Ex. 1002, 298, 253; Reply 12 ("This prosecution does not evince the '***clear*** reliance on the preamble during prosecution' required to impart weight to the preamble."). Because Patent Owner has not directed us to evidence that shows reliance by Examiner on the preamble to distinguish the claim over prior art, we are not

[7] There is no way to sort which MI patient population benefits from colchicine treatment, therefore, treatment would be reasonably be applied to all MI patients. *See* Ex. 1001, 32:19–67 (showing that only a fraction of the patients in either colchicine or placebo cohort suffered a stroke after their initial MI event); *see* Tr. 28:14–40:5.

[8] Claim 29, renumbered claim 27 in the '063 patent, is an original claim meaning that the claim has not been amended during prosecution. *See* Ex. 1002, 6.

persuaded that the preamble in this case has been transformed into a claim limitation.

Furthermore, we agree with Petitioner that "even if the preamble were given weight, stroke reduction is [an] inherent" outcome of the method disclosed in the COLCOT protocol. Reply 12.

Dr. Ridker, Petitioner's expert, in his declaration explains that one of ordinary skill in the art would understand the preamble reciting "[a] method of reducing the risk of a stroke in a patient after having an MI" to describe a treatment goal. Ex. 1049 ¶¶ 104–108. "[I]ts goal without requiring actually reducing stoke risk in every patient who received the medication." *Id.* ¶ 105; Dr. Ridker recognizes that some patients may benefit from the treatment while recognizing that other patients do not. Hence, Dr. Ridker concludes that the recitation of the reduction of stroke is a treatment goal. *Id*. This is further supported by Patent Owner's counsel explaining that the treatment is "not necessarily going to reduce the stroke in every patient. You know, overall, the cohort, there might be a reduction of stroke, but not every patient is going to have a reduced risk. So there are some patients for whom it may not be reduced." Tr. 48, 39:10–13; *see also* Ex. 1001, 32:19–67 (showing that in the placebo group 0.8% of the patients suffered stroke while in the colchicine study group 0.2% of the patients had a stroke – supporting the position that not every patient may have a reduced risk or stroke).

If the body of the claim sets out the complete invention, and the preamble is not necessary to give "life, meaning and vitality" to the claim, "then the preamble is of no significance to claim construction because it cannot be said to constitute or explain a claim limitation." *Pitney Bowes, Inc. v. Hewlett–Packard Co.*, 182 F.3d 1298, 1305 (Fed.Cir.1999). Here, the body of claim 27 recites "initiating the administration of colchicine at a daily low dose to the [MI] patient within about 30 days of an MI." Ex. 1001, 42:34–36. The claim body, therefore, identifies the patient population that is to be treated with the low dose colchicine and the time frame in which to initiate treatment. The preamble does not narrow the patient population nor change the way colchicine is administered – all that is required by the claim is that colchicine is administered to patients that have suffered a MI within a particular time period after the MI event. The step of the method is preformed the same way whether or not the patient experiences a reduction in the risk of stroke. *See, e.g., In re Hirao,* 535 F.2d 67, 70 (CCPA 1976) ("the preamble merely recites the purpose of the process; the remainder of the claim . . . does not depend on the preamble for completeness, and the process steps are able to stand alone."); *see* Pet. 23 ("'reducing the risk of a stroke' does 'not result in a manipulative difference in the steps of the claim' and is [therefore] non-limiting" (citing *Bristol-Myers*, 246 F.3d at 1375–76; Ex. 1049 ¶¶ 104–108)); Reply 11–13.

We agree with Petitioner that Examiner's claim interpretation during prosecution is in error. The body of claim 27 defines a structurally complete invention by identifying the target population as any patient that has had a myocardial infarction (MI) and the onset of treatment using 0.5 mg colchicine within a 30-day time window. *See Rowe v. Dror*, 112 F.3d 473, 478 (Fed. Cir. 1997) ("[W]here a patentee defines a structurally complete invention in the claim body and uses the preamble only to state a purpose or intended use for the invention, the preamble is not a claim limitation."); *IMS Tech., Inc. v. Haas Automation, Inc.,* 206 F.3d 1422, 1434 (Fed. Cir. 2000) ("If the preamble adds no limitations to those in the body of the claim, the preamble is not itself a claim limitation and is irrelevant to proper construction of the claim."). Petitioner contends, and we agree, that the recitation of "reducing the risk of a stroke" in the preamble, does not result in a difference in the way colchicine is administered to the identified patient population. *See* Pet. 23.

We have reviewed the parties' contentions, and agree with Petitioner that the body of claim 27 describes a structurally complete invention. Reply 12 (the claim "body stands complete on its own without preamble antecedent basis"). The preamble of claim 27, therefore, is not limiting because it simply expresses the intended result of the positively recited method step within the body of the claim. *See* Pet. 23; Reply 11–13; Dec. 10–12.

3. "wherein" clauses

Claim 8 contains three "wherein" clauses. *See* Ex. 1001, 41:50–60. The first wherein clause "wherein the colchicine is administered without pre-loading the patient with a higher dose of colchicine before the administration at the daily low dose" defines of how the colchicine is administered to the patient population. The second wherein clause "wherein the cardiovascular event is resuscitated cardiac arrest, stroke, or urgent hospitalization for angina requiring coronary revascularization" defines the type of cardiovascular event encompassed by the claim. And the third wherein clause "wherein the patient is at a lower risk of a cardiovascular event, relative to a patient not being administered colchicine" simply asks for a comparison between a patient receiving colchicine with a similar situated patient not receiving the colchicine treatment, in essence this is a comparison between the treatment group and the placebo group.

Patent Owner argues that "Examiner explicitly relied on the 'wherein' clauses to distinguish the claimed methods over the prior art during prosecution . . . [and the clauses are thereby] material to the Examiner's patentability determination." PO Resp. 12 (citing *Allergan Sales, LLC v. Sandoz, Inc.*, 935 F.3d 1370, 1376 (Fed. Cir. 2019)).

A review of the prosecution history shows that the limitation "wherein the colchicine is administered without pre-loading the patient with a higher dose of colchicine before the administration at the daily low dose" was added by amendment after the initial office action by Examiner. Ex. 1002, 328. Examiner noted that "[t]here is no motivation [in the art] to delete the loading dose." Ex. 1002, 398 (citing Ex. 1028; Ex. 1040); Reply 15 ("examiner predicated allowance on the no-loading-dose limitation"). In addition, Examiner noted that the art "does not teach show suggest or make obvious the cardiovascular events: resuscitated cardiac arrest, stroke, or urgent hospitalization for angina requiring coronary revascularization." Ex. 1002, 383, 398–399. This second wherein clause was entered after Examiner issued a final office action. Ex. 1002, 383. Here, the record supports that Examiner explicitly relied on the *first and second "wherein" clauses* to distinguish the claimed method over the prior art during prosecution. *See* Ex. 1002, 398–399. Accordingly, following *Allergan Sales* we will treat the *first and second* wherein clauses as limiting. *Allergan Sales*, 935 F.3d at 1367.

Petitioner contends that the third "wherein clause" of claim 8 – "wherein the patient is at a lower risk of a cardiovascular event" – "'is only a statement of purpose and intended result' that 'does not result in a manipulative difference in the steps of the claim' and is thus a non-limiting intended result and does not add patentable weight to the claims." Pet. 23 (citing *Bristol-Myers*, 246 F.3d at 1375–76; *Syntex (U.S.A.) LLC v. Apotex, Inc.*, 407 F.3d 1371, 1378 (Fed. Cir. 2005); *Minton v. Nat'l Ass'n of Sec. Dealers, Inc.*, 336 F.3d 1373,1381 (Fed. Cir. 2003)).

Patent Owner disagrees. PO Resp. 12 (citing *Allergan Sales,* 935 F.3d at 1376).

With respect to the third wherein clause Examiner, noted that the applied art ***did teach*** that "[p]atients administered colchicine were at a lower risk of a cardiovascular event, such as MI, relative to a patient not being administered colchicine." Ex. 1002, 398 (citing Ex. 1028). This third wherein clause was not entered in response to any rejection made by Examiner but instead is a limitation found in original claim 7. Ex. 1002, 383. Therefore, Examiner did not rely on the third wherein clause – "wherein the patient is at a lower risk of a cardiovascular event, relative to a patient not being administered colchicine" – to distinguish the claims over the prior art. Thereby, indicating that this third "wherein" clause was *not material* to Examiner's patentability determination. Accordingly, we decline Patent Owner's request that we treat the third "wherein" clause as material and limiting.

"An intended use or purpose usually will not limit the scope of the claim because such statements usually do no more than define a context in which the invention operates." *See Boehringer Ingelheim Vetmedica, Inc. v. Schering-Plough Corp.,* 320 F.3d 1339, 1345 (Fed. Cir. 2003). Although "[s]uch statements often . . . appear in the claim's preamble," a statement of intended use or purpose can appear elsewhere in a claim. *In re Stencel*, 828 F.2d 751, 754 (Fed. Cir. 1987).

We have reviewed the parties' contentions, and determine that Petitioner has provided a requisite showing that the third "wherein" clause is not limiting. Here, the third "wherein" clause of claim 8 merely describes the benefit of the administration of colchicine to the identified patient population. This "wherein clause," therefore, merely defines the result of the positively recited method steps. *See* Pet. 23; *see, e.g.*, *Minton*, 336 F.3d at 1381 (a "whereby clause in a method claim is not given weight when it simply expresses the intended result of a process step positively recited."). Accordingly, we determine that the third wherein clause – "wherein the patient is at a lower risk of a cardiovascular event, relative to a patient not being administered colchicine" – is not limiting and merely describes the intended result of administering colchicine to the defined patient population as described in the claim.

B. Level of Ordinary Skill in the Art

Petitioner contends a person of ordinary skill in the art ("POSA"), or a "skilled cardiologist," would have:

- An M.D. degree and be board-certified with a subspecialty in the treatment of cardiovascular disease.
- At least 3-5 years of experience, following board certification, in cardiology and cardiovascular disease treatment and sufficient clinical experience to understand standard good medical practice in patient management.
- Been familiar with acute coronary syndromes, stable coronary diseases, cardiovascular diseases, and their standard treatments, including the acute management and long-term secondary preventions of myocardial infarction.

- Been aware of research relating to cardiovascular disease progression, including the role of inflammatory pathways in secondary cardiovascular events.

Pet. 21 (citing Ex. 1049 ¶ 89).

Patent Owner's expert, Dr. Ardehali, disputes Petitioner's definition of a POSA. Ex. 2001 ¶ 12. Dr. Ardehali posits that Petitioner's definition is too narrow and that "a POSA for the '063 patent would be an M.D. with training and 3-5 years of experience in treating cardiovascular disease." *Id.* Although Dr. Ardehali disagrees with Petitioner's definition of a POSA, Dr. Ardehali states that his opinions remain the same regardless of whether the Board adopts Petitioner's definition or Patent Owner's definition. *Id.*

Based on the record, we adopt Petitioner's definition of a POSA. We further note that the prior art itself demonstrates the level of skill in the art at the time of the invention. *See Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (explaining that specific findings regarding ordinary skill level are not required "where the prior art itself reflects an appropriate level and a need for testimony is not shown") (quoting *Litton Indus. Prods., Inc. v. Solid State Sys. Corp.*, 755 F.2d 158, 163 (Fed. Cir. 1985)).

C. Anticipation by the COLCOT Protocol

Petitioner contends that claims 27, 29–36, and 39 are unpatentable as anticipated over the COLCOT protocol. Pet. 29–34; Reply 11–12. Patent Owner disagrees. PO Resp. 19–30. Patent Owner contends that the COLCOT protocol does not disclose the reduction of risk of stroke and that COLCOT is not prior art. PO Resp. 19–30; Sur-reply 12–16.

1. Overview of the COLCOT Protocol (Ex. 1006)

The Colchicine Cardiovascular Outcomes Trial (COLCOT) protocol is a copy of a webpage from ClinicalTrials.gov. Ex. 1006, 1.

According to the COLCOT protocol, "[a]therosclerosis is the most common cause of myocardial infarction, stroke and peripheral arterial disease. Research has clearly demonstrated that inflammation plays a key role in the initiation, progression and manifestations of atherosclerosis." *Id.* at 4. The COLCOT protocol explains that "[c]olchicine is an inexpensive, yet potent, anti-inflammatory drug approved for acute use in patients with gout and chronic use in patients with Familial Mediterranean Fever" and "[c]onsiderable work has highlighted the potential of colchicine in the treatment of cardiovascular diseases mediated by pro-inflammatory processes," such as coronary artery disease (CAD). *Id.*

The COLCOT protocol explains that "[t]he study evaluates whether long-term treatment with colchicine reduces rates of cardiovascular events in patients after myocardial infarction" (MI). *Id.* at 1. In the study, patients who suffered an acute MI within the last 30 days received either 0.5 mg/day colchicine or a matching placebo for about 2 years or until "the target of 301 primary endpoints has been reached." *Id.* The primary outcome measures in the study include cardiovascular death, resuscitated cardiac arrest, acute myocardial infarction, stroke, and hospitalization for angina requiring coronary revascularization. *Id.* at 2.

2. Analysis

a. *Claim 27*

Petitioner contends that the COLCOT protocol anticipates claim 27. Pet. 29–32; Reply 11–12. Patent Owner opposes. PO Resp. 19–30; Sur-reply 12–16.

i. Prior art status of the COLCOT protocol

Mr. Cruitt is a research librarian and provided testimony with respect to the internet archive service known as the Wayback Machine (https://web.archive.org/) and the retrieval of the COLCOT protocol from the internet archive. *See* Ex. 1038 ¶¶ 1–4, 15–19, 23–25. Patent Owner contends that Mr. Cruitt "is not an objective expert witness, and his biased opinion should be stricken and afforded no weight." PO Resp. 20 (citing Paper 2; Ex. 1038 ¶ 2). Patent Owner basis this assertion on Mr. Cruitt's relationship with Petitioner's original law firm that filed the *inter partes* review petition. *Id.* Patent Owner further contends that according to the FDA's "record history reports the same or similar data regarding the timing of certain submissions and filings, however the public availability of the COLCOT protocol is listed as July 16, 2020." *Id.* (citing Ex. 2069, 13). In other words, Patent Owner contests the public availability of Exhibit 1006 – the COLCOT protocol.

Further calling into question Mr. Cruitt's analysis Patent Owner contends that Exhibit 1006 "is the linked version of the study protocol, which is internally dated November 15, 2018, more than eighteen months *after* Mr. Cruitt alleges the protocol was publicly available." PO Resp. 20. Patent Owner contends that "Petitioner has still failed to meet its burden to show that EX1006 is publicly available prior art. Mr. Cruitt agrees that

EX2069 refers to a July 16, 2020 upload date for the study protocol." Sur-reply 13 (citing Ex. 1072 ¶ 5).

Mr. Cruitt testified that "a version of the COLCOT Protocol was submitted on March 6, 2017 and publicly accessible by March 8, 2017." Ex. 1072 ¶ 5; Ex. 1038 ¶¶ 15–19, 23–25. In response to Patent Owner's assertion that the COLCOT protocol listed July 16, 2020 as the upload date (*see* PO Resp. 20), Mr. Cruitt explains that "pages 1-2 of Exhibit 2069 list ten version[s] of the COLCOT study record submitted from September 15, 2015 through September 23, 2020. As I explained in detail in my First Declaration, a version of the COLCOT Protocol was submitted on March 6, 2017 and publicly accessible by March 8, 2017." Ex. 1072 ¶ 5.

In response to Patent Owner's questioning the public availability of Exhibit 1006, Petitioner provided a declaration from a second librarian. Reply 10; PO Resp. 20 (Patent Owner asserting that the first librarian "Mr. Cruitt's declaration [is] not reliable due to bias as a result of fiduciary duty owed to his employer—Petitioner's law firm"). Dr. Mullins, the second librarian, confirms that Exhibit 1006 – the COLCOT protocol – was publicly accessible by March 2017. Reply 10 (citing Ex. 1073 ¶¶ 22–29). In the declaration Dr. Mullins explains that "[t]he Internet Archive is a digital archive/library that began in 1996 to 'scrape' publicly accessible webpages. . . . It is more or less a 'snapshot' of the website on that date, further information is available at this URL: https://archive.org/about/." Ex. 1073 ¶ 16. Dr. Mullins testifies "that COLCOT, Exhibit 1006, is an authentic document, and was publicly accessible and documented by the evidence in Attachment A-1, Wayback Machine, October 20, 2017." *Id.* at ¶ 31.

Below is a snapshot of the screen capture header found in Exhibit 1006:

Ex. 1006, 1.

Below is a snapshot of the header from Dr. Mullins Attachment -A1:

Screen capture created by J L Mullins from Wayback Machine, Internet Archive, on June 24, 2024, at URL:
https://web.archive.org/web/20171020004013/https:/clinicaltrials.gov/ct2/show/NCT02551094

Ex. 1073, 31.

Dr. Mullins testified that a comparison of "the screen capture I obtained from the Wayback Machine, the webpage archived on October 20, 2017, which is Attachment A-1, with Exhibit 1006, and the content is an exact match." Ex. 1073 ¶ 21. Dr. Mullins further testifies that "[t]o verify authenticity and public access, a citation to a publication is evidence that the publication (article, report, thesis, etc.) is considered authoritative and was publicly accessible as of the date of the published citation." *Id.* ¶ 22. Dr. Mullins retrieved two articles that each cite accessing the COLCOT protocol through the FDA website at the following URL: https://www.clinicaltrials.gov/ ct2/show/NCT02551094?term-colcot&rank-1. The two different articles are dated October 10, 2017 and August 30, 2018 establishing that the COLCOT protocol was not only known but also publicly accessible to persons of ordinary skill in the art. *Id.* ¶¶ 23–28, 31. Based on these disclosures, Dr. Mullins concludes the Mr. Cruitt's initial determination that the COLCOT protocol document was at least available as of October 20, 2017 is correct. *Id.* ¶ 32.

The totality of the evidence provided by Petitioner persuades us that Exhibit 1006 was publicly accessible more than one year prior to earliest filing date of Nov. 15, 2019 of the ’063 patent. Reply 10–11; Ex. 1001, code (60). Exhibit 1006 appears to be identical to the document retrieved by Dr. Mullins using the same URL. *Compare* Ex. 1006 *with* Ex. 1073, 31–35 (attachment A-1). In addition, Dr. Mullins directed us to two contemporaneous articles that identify the same COLCOT protocol associated with the same URL. *See* Ex. 1073 ¶¶ 23–28 (attachment A–2 and A–3). Based on these disclosures, we determine that the totality of the evidence corroborates the testimony of Mr. Cruitt (Ex. 1038; Ex. 1072) or Dr. Mullins (Ex. 1073) and is sufficient to show that Exhibit 1006 was publicly accessible more than one year prior to the filing date of the ’063 patent.

ii. Colchicine administration

Patent Owner contends that the COLCOT protocol “does not disclose a risk reduction of stroke.” PO Resp. 19, 22. Patent Owner contends that “COLCOT merely tested a ‘*hypothesis* that treatment of the underlying inflammatory process will contribute to improved cardiovascular clinical outcomes.’” PO Resp. 22. Patent Owner further contends that the COLCOT protocol does not inherently disclose a reduction in the risk of stroke. PO Resp. 23. “COLCOT does not meet this strict standard for inherent anticipation. COLCOT merely provides an invitation to perform the clinical study.” *Id.* at 24.

Patent Owner contends that Petitioner’s reliance on *In re Montgomery* fails because Patent Owner in this case relies on contemporaneous rebuttal evidence to show “low-dose colchicine does not reduce the risk of recurrent

CV events, including stroke, in patients post-PCI." PO Resp. 25 (citing *In re Montgomery*, 677 F.3d 1375, 1380 (Fed. Cir. 2012) citing *Metabolite Labs. Inc. v. Lab. Corp. of Am. Holdings*, 370 F.3d 1354, 1367 (Fed. Cir. 2004) ("[a]n invitation to investigate is not an inherent disclosure")). Patent Owner contends that the

> COPS [study published online August 29, 2020] investigated whether patients with acute coronary syndromes ('ACS') could be effectively treated with colchicine in addition to standard secondary therapies. . . . [The study concluded that] the addition of low-dose oral colchicine to standard medical therapy during hospitalization and continued for 12 months did not affect the rate of the primary composite outcome of death, ACS, ischemia-driven urgent revascularization, and stroke compared with standard medical therapy alone.

PO Resp 25–26 (citing Ex. 2056).

Patent Owner contends that Petitioner's own expert testifies the he would not rely on the COLCOT protocol. PO Resp. 19. Additionally, Patent Owner contends that "Petitioner's expert, Dr. Ridker himself undermined Petitioner's ability to rely on COLCOT to show invalidity" based on anticipation. PO Resp. 20. Patent Owner contends that Dr. Ridker testified that he did not recall seeing the COLCOT protocol and that "the full protocol is what he would rely upon to understand a trial." *Id.* at 21 (citing Ex. 2071, 139:3–140:5, 140:8–141:15, 146:11–147:11, 151:9–152:22; Ex. 2069).

Based upon our review of the parties' arguments and supporting evidence in the record, we determine that Petitioner has demonstrated by a preponderance of evidence that claim 27 is anticipated by COLCOT protocol. We address Patent Owner's contentions below:

a) Reducing stroke risk

In the COLCOT protocol, the target patient population are patients that "have suffered a documented acute myocardial infarction within the last 30 days." Ex. 1006, 1; Pet. 30–32; Ex. 1049 ¶¶ 125–132. In other words, patients that suffered a myocardial infarction *more than 30 days ago* are excluded from the study. The COLCOT protocol, therefore, includes patients that had a documented myocardial infarction, but that the myocardial infarction event was less than 30 days from the start of treatment, as required in claim 27. *See* Ex. 1001, 42:35–36.

The COLCOT protocol instructs to sort patients into either a treatment or placebo group. Ex. 1006, 1. The treatment group will receive "colchicine (0.5 mg per day) . . . for an estimated 2 years period." Ex. 1006, 1; Pet. 30–32; Ex. 1049 ¶¶ 125–132. The treatment group in the COLCOT protocol, therefore, meets the limitation of claim 27 that administers "low dose" colchicine to patients. *See* Ex. 1001, 42:36, *see also id.* at 42:50–53 (describing that colchicine is administered at 0.5 mg once per day).

For the reason discussed above (*see* II.A.2), we find that the preamble of claim 27 is not limiting because it simply expresses the intended result of the positively recited method step within the body of the claim. The step of the method is performed the same way whether or not the patient experiences a reduction in the risk of stroke. *See, e.g., In re Hirao,* 535 F.2d at 70 ("the preamble merely recites the purpose of the process; the remainder of the claim . . . does not depend on the preamble for completeness, and the process steps are able to stand alone."); *see also Bristol-Myers*, 246 F.3d at 1375–76 ("the expression '[a] method for treating a cancer patient to effect regression of a taxol-sensitive tumor, said method being associated with reduced hematologic toxicity' in the preambles . . . is only a statement of purpose and intended result. The expression does not result in a manipulative difference in the steps of the claim.").

Furthermore, by treating the same patient population with the same drug – low dose colchicine – we agree with Petitioner that the result of lowering the risk of stroke is an inherent feature of applying the treatment method taught in the COLCOT protocol. *See Bristol-Myers Squibb Co.*, 246 F.3d at 1376 ("Newly discovered results of known processes directed to the same purpose are not patentable because such results are inherent."

Thus, we are persuaded that Petitioner has demonstrated by a preponderance of evidence that the COLCOT protocol teaches administering 0.5 mg per day colchicine to patients that suffered a myocardial infarction within the last 30 days.

b) Contemporaneous rebuttal evidence

Patent Owner contends that the "COLCOT [protocol] does not meet this strict standard for inherent anticipation. COLCOT merely provides an invitation to perform the clinical study." PO Resp. 24 (citing *In re Montgomery*, 677 F.3d 1375, 1380 (Fed. Cir. 2012) citing *Metabolite Labs. Inc. v. Lab. Corp. of Am. Holdings*, 370 F.3d 1354, 1367 (Fed. Cir. 2004) ("[a]n invitation to investigate is not an inherent disclosure")). Patent Owner contends that we should consider the COPS (Colchicine in Patients With Acute Coronary Syndromes) study, a post filing date study that published November 17, 2020, to arrive at the conclusion that the COLCOT protocol does not show a reduction in the risk of stroke. *See id.* at 24–26 (citing Ex. 2056).

Petitioner responds, and we agree, that Patent Owner "advances a policy argument not based on the law and relying on a ***dissent*** as its only support. [PO] Response 27-28 (citing *In re Montgomery*, 677 F.3d 1375, 1383-84 (Fed. Cir. 2012) (Lourie, J., dissenting)). The *Montgomery* majority held similar method-of-treatment claims anticipated by a trial protocol, as the Board should do here. *Id.* at 1377-78, 1381 (disclosure of protocol inherently treats or prevents stroke.)." Reply 13.

In *Montgomery* our reviewing Court explained:

> We have repeatedly held that "[n]ewly discovered results of known processes directed to the same purpose are not patentable because such results are inherent." *Bristol–Myers Squibb,* 246 F.3d at 1376. As we stated in *Cruciferous Sprout,* 301 F.3d at 1350, "[i]t matters not that those of ordinary skill heretofore may not have recognized the[] inherent characteristics of the [prior art]."

Montgomery, 677 F.3d at 1381. Furthermore, "finding of anticipation 'does not require the actual creation or reduction to practice of the prior art subject matter; anticipation requires only an enabling disclosure.'" *Arbutus Biopharma Corp. v. ModernaTX, Inc.*, 65 F.4th 656, 662 (Fed. Cir. 2023) (citing *Schering Corp. v. Geneva Pharms.*, 339 F.3d 1373, 1380 (Fed. Cir. 2003)).

As discussed above, the COLCOT protocol recites the following:

> [p]atients who have suffered a documented acute myocardial infarction within the last 30 days, are treated according to the national guidelines and after having completed any planned percutaneous revascularization procedures associated with their initial infarction will receive either colchicine (0.5 mg per day)

> . . . for an estimated 2 years period or until the target of 301 primary endpoints has been reached.

Ex. 1006, 1. Here, the COLCOT protocol treats the same patient population with the same drug – low dose colchicine – efficacy is inherent in carrying out the protocol disclosed in the art. *See Montgomery*, 677 F.3d at 1381.

"[P]roof of efficacy [,however,] is not required for a prior art reference to be enabling for purposes of anticipation . . . [T]he proper issue is whether the [prior art] is enabling in the sense that it describes the claimed invention sufficiently to enable a person of ordinary skill in the art to carry out the invention." *Impax Labs., Inc. v. Aventis Pharms. Inc.*, 468 F.3d 1366, 1383 (Fed. Cir. 2006). Here, the COLCOT protocol tells the person of ordinary skill in the art what patient population to treat, when to start treatment, the duration of treatment, and how much colchicine to administer. *See* Ex. 1006, 1; Reply 11 (Patent Owner's expert Dr. Ardehali "admitted EX1006 teaches effective stroke treatment within about 30 days of an MI. EX1076, 119:20-121:8 ('So in COLCOT … if it is started within the first 30 days … it would reduce the risk of stroke in patients after MI.')); Ex. 1005 ¶¶ 124, 126–131; Reply 11–13. We agree with Petitioner that the COLCOT protocol provides an enabling disclosure as recognized by persons of ordinary skill in the art.

Additionally, we are not persuaded Patent Owner's contention that there is unpredictability in applying colchicine treatment to cardiovascular patients. PO Resp. 25–26 (citing Ex. 2056 (COPS trial), 2070 (COPS trial online)). Dr. Ridker explains that the COLCOT protocol "is a trial in stable patients. The studies in the acute setting [disclosed in COPS trial] are not informative about effectiveness in treating chronic atherosclerosis in stable patients." Ex. 1075 ¶ 17. While the COPS study is a trial "conducted in the setting of acute ischemia or acute coronary syndromes" COLCOT patients have stable coronary artery disease. *Id.* ¶ 22; Ex. 2056, 2 (describes that the COPS trial relied on patients with acute coronary syndrome (ACS) "defined as symptoms of acute myocardial ischemia associated with either elevated troponin or ECG changes, which included STEMI [ST-segment–elevation myocardial infarction], non-STEMI, and unstable angina" the reference acknowledges that COLCOT and LoDoCo trial "patients with ACS and stable CAD.") We credit Dr. Ridker's testimony that the studies cited by Patent Owner as being unpredictable or ineffective are not persuasive because they were conducted with an unstable patient population which is a different patient population to the patients in the COLCOT protocol. Ex. 1075 ¶ 6.

Dr. Ridker further testifies that

> colchicine at a dose of 0.5 mg once daily had already been established as a highly effective method to reduce cardiovascular event rates in patients with or at high risk for chronic atherosclerotic disease, as well established in three large scale randomized clinical trials: LoDoCo, LoDoCo2 and COLCOT, the latter two of which were additionally placebo-controlled.

Ex. 1075 ¶ 6. This testimony is further supported in the regulatory filings by Patent Owner. *Id.* (citing Ex. 1051–1053, Ex. 1074); *see* Ex. 1074, 18 ("The characteristics of patients in LODOCO and LODOCO2 were well representative of the stable CAD disease").

Thus, we are not persuaded by Patent Owner's contention that contemporaneous evidence supports the position that the COLCOT protocol is not an enabling disclosure as recognized by persons of ordinary skill in the art.

c) Full-length protocol

Patent Owner contends that we should "disregard Dr. Ridker's testimony regarding [anticipation based on the COLCOT protocol], because that testimony is a repetition of attorney argument." PO Resp. 21 ("At deposition Dr. Ridker testified under oath that he did not recall ever seeing COLCOT. EX2071 at 139:3-140:5, 140:8-141:15"). Specifically, Patent Owner contends that "Dr. Ridker further testified that he would not look to a clintrials.gov summary reference to understand a trial's protocol, and instead would rely on the entire protocol as submitted to the FDA or a publication." *Id.* (citing Ex. 2071, 139:3-140:5, 140:8-141:15, 146:11-147:11.).

Petitioner responds that Dr. Ridker's not being able to immediately recall Exhibit 1006 during deposition does not mean that he did not persuasively testify in his declarations. Reply 11 (citing Ex. 1075 ¶¶ 35–46; Ex. 1049 ¶¶ 123–129). In his second declaration, Dr. Ridker further explains that "[w]hile I would rely on the complete protocol for scientific purposes, the brief summary suffices for analyzing patentability because the brief summary includes the pertinent information." Ex. 1075 ¶ 41.

A "finding of anticipation 'does not require the actual creation or reduction to practice of the prior art subject matter; anticipation requires only an enabling disclosure.'" *Arbutus Biopharma Corp. v. ModernaTX, Inc.*, 65 F.4th 656, 662 (Fed. Cir. 2023) (citing *Schering Corp. v. Geneva Pharms.*, 339 F.3d 1373, 1380 (Fed. Cir. 2003)). Here, Dr. Ridker explained that "[i]t was not necessary to refer to the long-form trial protocol for the purpose of my first declaration because the clinicaltrials.gov reference includes all relevant details of the treatment method used in the clinical trial, including a description of the background, the purpose, the dose, dosage regime, patient inclusion and exclusion criteria, outcomes, control group, and duration of the trial." Ex. 1075 ¶ 38.

Accordingly, we are not persuaded that Patent Owner's contention that the COLCOT protocol provides insufficient information to carry out the methods disclosed.

iii. Summary claim 27

For the reasons discussed above, we determine that Petitioner has demonstrated by a preponderance of evidence that claim 27 is anticipated by the COLCOT protocol.

b. *Claims 29–32 (Additional Medication)*

Claim 29 depends from claim 27 and further recites "wherein the patient was prescribed a medication." Ex. 1001, 42:40–41. Claim 30 depends from claim 29 and ultimately claim 27 and further recites "wherein the medication is an antiplatelet agent." Ex. 1001, 42:42–43. Claim 31 depends from claim 29 and ultimately claim 27 and further recites "wherein the medication is aspirin." Ex. 1001, 42:44–45. Claim 32 depends from claim 29 and ultimately claim 27 and further recites "wherein the medication is a statin." Ex. 1001, 42:46–47.

The COLCOT protocol recites that a "[p]atient must be treated according to national guidelines (including anti-platelet therapy, statin, renin-angiotensin-aldosterone system inhibitor (preferably angiotensin-converting enzyme) and beta-blocker when indicated)." Ex. 1006, 4. "[A] POSA would have immediately envisaged aspirin—the most common cardiovascular antiplatelet therapy as of 2019—from COLCOT Protocol's disclosure of "antiplatelet therapy" according to "national guidelines." Pet. 33; Ex. 1049 ¶ 135; *see, e.g.*, Ex. 1021, 3 ("A dose of 160 to 325 mg [aspirin] should be given on day 1 of AMI and continued indefinitely on a daily basis"); Ex. 1009, 31 ("For long-term prevention, low aspirin doses (75–100mg) are indicated").

Based on the disclosure in the COLCOT protocol and for the reasons stated in the Petition, which we adopt as our own findings, we are persuaded that Petitioner has demonstrated by a preponderance of evidence that claims 29–32 are anticipated by the COLCOT protocol. *See* Pet. 32–33.

c. *Claims 33, 39 (Patient Population)*

Claim 33 depends from claim 27 and further recites "wherein the patient has atherosclerotic coronary artery disease." Ex. 1001, 42:48–49. Claim 39 depends from claim 27 and further recites "wherein the patient is an adult human." Ex. 1001, 42:62–63.

The COLCOT protocol recites that "[a]therosclerosis is the most common cause of myocardial infarction, stroke and peripheral arterial disease." Ex. 1006, 4; Ex. 1049 ¶ 136 ("A skilled cardiologist would have understood that this teaching in COLCOT Protocol is consistent with the prior understanding that the vast majority of MI patients, including those treated by COLCOT Protocol, would have atherosclerotic coronary artery disease.").

The COLCOT protocol further recites that the ages of patients eligible for the study 18 years and older (Adult, Senior). Ex. 1006, 4. "Males and females of at least 18 years of age capable and willing to provide informed consent." *Id.*; Ex. 1049 ¶ 137 ("A skilled cardiologist would have understood that such specific patient population would be adult human patients”). Additionally, the requirement for providing informed consent in the COLCOT protocol further support the position that the patients are human patients.

We are persuaded, for the reasons stated in the Petition, which we adopt as our own findings, that Petitioner has demonstrated by a preponderance of evidence that claims 33 and 39 are anticipated by the COLCOT protocol. *See* Pet. 33.

d. *Claims 34, 35 (Dose and Frequency)*

Claim 34 depends from claim 27 and further recites “wherein the co[l]chicine is administered at 0.5 mg.” Ex. 1001, 42:50–51. Claim 35 depends from claim 27 and further recites “wherein the colchicine is administered once per day.” Ex. 1001, 42:52–53.

The COLCOT protocol recites administering “0.5 mg tablet of colchicine taken once a day.” Ex. 1006, 3.

Based on this disclosure in COLCOT and for the reasons stated in the Petition, which we adopt as our own findings, we are persuaded that Petitioner has demonstrated by a preponderance of evidence that claims 34 and 35 are anticipated by the COLCOT protocol. *See* Pet. 34.

e. *Claim 36 (Initiation Upon Assessment)*

Claim 36 depends from claim 27 and further recites “wherein the administration of colchicine is initiated upon assessment in (a) an emergency department (ED), (b) the hospital, or (c) a medical office setting.” Ex. 1001, 42:54–57.

The COLCOT protocol recites that the location for the trial is the “Montreal Heart Institute.” Ex. 1006, 6; Ex. 1049 ¶ 139 (“the ‘Montreal

Heart Institute,' which was a well-known hospital facility"). The COLCOT protocol recites that "Patient is considered by [t]he investigator, for any reason, to be an unsuitable candidate for the study." Ex. 1006, 5. We credit Dr. Ridker's testimony that a cardiologist tends to work in a medical setting such as an office. Ex. 1049 ¶ 139 ("A skilled cardiologist would have understood that the assessment by an investigator would occur at a hospital (the study site), emergency department, or medical office setting").

We are persuaded, for the reasons stated in the Petition, which we adopt as our own findings, that Petitioner has demonstrated by a preponderance of evidence that claim 36 is anticipated by the COLCOT protocol. *See* Pet. 34.

f. *Summary*

For the reasons discussed above, we determine that Petitioner has demonstrated by a preponderance of evidence that claims 27, 29–36, and 39 are is anticipated by the COLCOT protocol.

D. *Obviousness over Nidorf, COLCOT Protocol, ESC Guidelines, and COLCRYS*

Petitioner contends that claims 1–50 are unpatentable as obvious over Nidorf, the COLCOT protocol, the ESC guidelines, and COLCRYS. Pet. 34–54. Patent Owner opposes. PO Resp. 30–55.

1. *Overview of Nidorf (Ex. 1007)*

Nidorf is titled "Method of Treating Cardiovascular Events Using Colchicine Concurrently with an Antiplatelet Agent" and issued on February 19, 2019. Ex. 1007, codes (45), (54). Nidorf relates to "[m]ethods of treating and/or preventing a cardiovascular event in a patient, the method comprising orally administering a colchicine to a patient who is receiving concurrent treatment with at least one antiplatelet agent." *Id.* at code (57). According to Nidorf, "[c]olchicine is an anti-inflammatory drug with a long history in human medicine, used for the symptomatic treatment of inflammatory diseases, most prominently gout," but also familial Mediterranean fever and Behçet's disease. *Id.* at 6:19–21; *see id.* at 1:19–28. Colchicine has also been proposed to be effective in therapy for cardiovascular diseases such as recurrent and acute pericarditis, post-pericardiotomy syndrome (PPS), and atherosclerotic vascular disease. *Id.* at 1:29–2:33.

Nidorf's method includes treating and/or preventing a cardiovascular event in a patient, such as acute coronary syndrome, out-of-hospital cardiac arrest, and/or noncardioembolic ischemic stroke, by administering a composition comprising an effective amount of colchicine to a patient who is receiving concurrent treatment with at least one antiplatelet agent, such as aspirin. *Id.* at 2:56–3:6. In some embodiments, the amount of colchicine in the composition is 0.5 mg or 0.6 mg, but the amount of colchicine "will vary depending upon the host treated and the particular mode of administration." *Id.* at 14:46–49; *see id.* at 2:56–3:6.

Nidorf's composition may be administered orally, parenterally, by inhalation, or topically as a single dose, multiple doses, or over an established period of time, and the dosage regimen can be adjusted to provide the desired therapeutic or prophylactic response. *Id.* at 13:55–57, 14:49–53. Nidorf states that its composition can be administered with a single dose or up to 10 doses, given once daily or up to 10 times per day, but preferably once per day as a single dose. *Id.* at 15:31–38. Nidorf also explains that its composition can be administered regularly for long periods of time, including from a period of weeks to up to 10 years, depending on the severity of disease, the overall health of the patient, any additional medications the patient is taking, and whether the treatment is prophylactic or not. *Id.* at 15:39–58.

2. *Overview of the European Society of Cardiology (ESC) Guidelines (Ex. 1009)*

The "2017 ESC Guidelines for the management of acute myocardial infarction in patients presenting with ST-segment elevation." Ex. 1009, 1 (Title). The ESC guidelines defines acute myocardial infarction (AMI) as a condition with evidence of myocardial injury (i.e., "an elevation of cardiac troponin values with at least one value above the 99th percentile upper reference limit") with necrosis consistent with myocardial ischaemia. *Id.* at 6. The ESC guidelines provide treatment and management guidance for emergency care of AMI and hospitalization, reperfusion therapy, post-discharge care, and complications. *Id.* at 2–3. The ESC guidelines also discusses how to assess the quality of care and areas for future research. *Id.* The ESC guidelines was developed by selected experts in the field who "undertook a comprehensive review of the published evidence for management of [AMI] according to ESC Committee for Practice Guidelines (CPG) policy" and "[a] critical evaluation of diagnostic and therapeutic procedures was performed, including assessment of the risk–benefit ratio." *Id.* at 5.

3. *Overview of COLCRYS (Ex. 1016)*

COLCRYS is a package insert from "COLCRYS™ (colchicine, USP) tablets for Oral use." Ex. 1016, 1. COLCRYS states that the tablets are indicated for use to treat gout flares and Familial Mediterranean fever (FMF). *Id.* COLCRYS may be given on a long-term basis for treating FMF, but "the safety and efficacy of repeat treatment in gout flares has not been evaluated." *Id.* at 2 (emphasis omitted). It is thought that COLCRYS works by "interfer[ing] with the intracellular assembly of the inflammasome complex present in neutrophils and monocytes that mediates activation of interleukin-1β." *Id.* at 9.

To treat gout flares, COLCRYS should be administered at a dosage of "1.2 mg (2 tablets) at the first sign of a gout flare followed by 0.6 mg (1 tablet) one hour later." *Id.* at 1. "The maximum recommended dose for treatment of gout flares is 1.8 mg over a 1 hour period." *Id.* at 2.

To treat FMF, COLCRYS should be administered at the total daily dose in one or two divided doses and increased or decreased "as indicated and as tolerated in increments of 0.3 mg/day, not to exceed the maximum recommended daily dose." *Id.* at 1. For adults and children older than 12 years, the daily dosage is 1.2–2.4 mg. *Id.* For children 6 to 12 years, the daily dosage is 0.9–1.8 mg. *Id.* For children 4 to 6 years, the daily dosage is 0.3–1.8 mg. *Id.*

4. Analysis

A patent claim is unpatentable under 35 U.S.C. § 103 if the differences between the claimed subject matter and the prior art are such that the subject matter, as a whole, would have been obvious at the time the invention was made to a person having ordinary skill in the art to which the subject matter pertains. *KSR Int'l Co. v. Teleflex Inc.*, 550 U.S. 398, 406 (2007). "An obviousness determination requires finding both 'that a skilled artisan would have been motivated to combine the teachings of the prior art references to achieve the claimed invention, and that the skilled artisan would have had a reasonable expectation of success in doing so.'" *CRFD*

Research, Inc. v. Matal, 876 F.3d 1330, 1340 (Fed. Cir. 2017) (quoting *Intelligent Bio-Sys., Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367–1368 (Fed. Cir. 2016)).

a. *Claim 27*

Petitioner contends that claim 27 is unpatentable as obvious over Nidorf, COLCOT protocol, ESC guidelines, and COLCRYS. Pet. 34–35, 44–46. Patent Owner opposes. PO Resp. 48–51.

For the reasons discussed above with respect to claim 27, we determine that Petitioner has shown by a preponderance of the evidence that claim 27 is anticipated by the COLCOT protocol. *See above* II.C.2.a; *see Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1373 (Fed. Cir. 2019) ("[I]t is well settled that a disclosure that anticipates under § 102 also renders the claim invalid under § 103, for anticipation is the epitome of obviousness.") (internal quotations and citations omitted).

We are persuaded, for the reasons stated above (see II.C) and as stated in the Petition, which we adopt as our own findings, that Petitioner has demonstrated by a preponderance of evidence that claim 27 would have been obvious. *See* Pet. 44–46.

i. Claim 28

Claim 28 depends from claim 27 and further recites "wherein the method comprises administering co[l]chicine within 4 to 7 days of the MI." Ex. 1001, 42:37–39.

Petitioner contends that claim 28 would have been obvious over Nidorf, the COLCOT protocol, the ESC guidelines, and COLCRYS. Pet. 44–46, 53–54. Specifically, Petitioner contends that "[t]here would have been a specific motivation and a reasonable expectation of success to initiate colchicine treatment post-MI during follow-up care, e.g., within days 4-7 post-MI, if the treatment had not already been started during hospitalization." Pet. 53.

Patent Owner contends that the art would have taught away from initiating treatment so early after MI. PO Resp. 55 (citing Ex. 2009). Patent Owner contends that "O'Keefe concluded that low-dose colchicine initiated within one day of coronary angioplasty 'proved ineffective in preventing restenosis after coronary angioplasty.'" *Id.* at 44 (citing Ex. 2009, 2). Specifically, O'Keefe taught that administering 0.6 mg colchicine twice a day to patients starting either 12 hours before or 24 hours after angioplasty. Ex. 2009, Abstract. Treatment was continued for 6 months until angiographic follow up. Ex. 2009, 2.

We are not persuaded by Patent Owner's contention that O'Keefe teaches away from administering colchicine early after an MI. As Dr. Ridker explains the study contained some flaws in that it relied on a small sample size and that it used "surrogate endpoints as primary outcome (such as change in hsCRP, flow-mediated dilation, or troponin I measurement) that may or may not ultimately prove informative when large hard outcomes trials are conducted." Ex. 1075 ¶ 19. Dr. Ridker further explains that surrogate end points can be considered informative but given the more recent studies showing "the effect of low-dose colchicine on hard clinical endpoints was already established by LoDoCo and LoDoCo2." *Id.*

In the COLCOT protocol, the target patient population are patients that "have suffered a documented acute myocardial infarction within the last 30 days." Ex. 1006, 1; Pet. 53–54; Ex. 1049 ¶¶ 210–216. The COLCOT protocol, therefore, includes patients that had a documented myocardial infarction, but that the myocardial infarction event occurred less than 30 days from the start of treatment. In other words, the COLCOT protocol encompasses a time range of 0–30 days in which to initiate colchicine treatment.

Petitioner contends that the ESC guidelines provide motivation to initiate colchicine treatment within 4–7 days post-MI (if not already initiated). Pet. 54 (citing Ex. 1009, 24; Ex. 1013, 1; Ex. 1049 ¶¶ 211–215). According to the ESC guidelines,

> A short hospital stay implies limited time for proper patient education and up-titration of secondary prevention treatments. Consequently, these patients should have early post-discharge consultations with a cardiologist, primary care physician, or specialized nurse scheduled and be rapidly enrolled in a formal rehabilitation programme, either in-hospital or on an outpatient basis.

Ex. 1009, 24.

We credit Dr. Ridker's testimony that "there had been clear reasons to initiate treatment during the typical 3-day post-MI hospital period, [but] it would have also been obvious to initiate colchicine treatment shortly after discharge, e.g., within days 4-7 post-MI, during follow-up care if the treatment had not already been started during hospitalization." Ex. 1049 ¶ 212. As Dr. Ridker further explains that early follow up with a patient after MI results in better long-term outcomes. *Id.* ¶ 214 (citing Ex. 1013, 1 ("See You in 7 Collaborative participation was associated with significantly lower 30-day readmissions and Medicare payments in [heart failure] HF patients."); Ex. 1009, 30).

We are persuaded, for the reasons stated in the Petition, which we adopt as our own findings, that Petitioner has demonstrated by a preponderance of evidence that claim 28 would have been obvious. *See* Pet. 53–54.

b. *Claim 1*

Petitioner contends that claim 1 is unpatentable as obvious over Nidorf, the COLCOT protocol, the ESC guidelines, and COLCRYS. Pet. 36–44. Patent Owner opposes. PO Resp. 30–55. Specifically, Patent Owner argues that the Petition fails to address: (a) the removal of a pre-loading dose; (b) applying the claimed method to patients who have already been provided a percutaneous intervention; (c) administration of colchicine within the first 3 days after an MI; and (d) there was a reduction in the risk of experiencing another cardiovascular event, including stroke, in patients who were treated with the claim method versus those who were not. *Id.* at 31.

i. *(Preamble 1) A method of treating a patient after having a myocardial infarction (MI)*

Petitioner contends that based on the teachings of Nidorf, the COLCOT protocol, the ESC guidelines, and COLCRYS, a person of ordinary skill in the art at the time the invention was made would have been motivated to start "secondary preventive therapy in post-MI coronary artery disease patients." Pet. 34 (citing Ex. 1007; Ex. 1006), 36 (citing Ex. 1007, 20:3–5; Ex. 1049, ¶¶ 68, 145–146).

Patent Owner opposes but does not articulate what limitation is missing based on combination of references as presented by Petitioner. *See generally* PO Resp.[9]

[9] Patent Owner throughout the Response repeatedly asserts that the COLCOT protocol is not prior art. *See generally* PO Resp. For the reason

Nidorf describes adding "0.5 mg/day of colchicine to standard secondary prevention therapies including aspirin and high dose statins reduces the risk of cardiovascular events in patients with objectively diagnosed and clinically stable coronary disease." Ex. 1007, 18:2–6.

The COLCOT protocol similarly describes treating "[p]atients who have suffered a documented acute myocardial infarction within the last 30 days" with colchicine at a dose of 0.5 mg/day. Ex. 1006, 1. The patients are otherwise "treated according to the national guidelines and after having completed any planned percutaneous revascularization procedures associated with their initial infarction." *Id.*

The ESC guidelines teach the management of patients having suffered acute myocardial infarctions. *See generally* Ex. 1009. The ESC guidelines include instructions for emergency care and reperfusion therapies. *See* Ex. 1009, 8–24. The ESG guidelines also include instructions for the administration of secondary therapies such as antithrombic therapies including aspirin, dual antiplatelet therapy, beta-blockers, lipid-lowering therapy, angiotensin-converting enzyme (ACE) inhibitors, angiotensin II receptor blockers, and mineralocorticoid/aldosterone receptor antagonists. *Id.* at 31–35.

Based upon our review of the parties' arguments and supporting evidence, we determine that Petitioner as demonstrated by a preponderance of evidence that the preamble of claim 1 would have been suggested by the combination of Nidorf, the COLCOT protocol, and the ESC guidelines.

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discussed above in section II.C.2.a.i, we are persuaded that Exhibit 1006 was publicly accessible before the filing date of Nov. 15, 2019 of the '063 patent.

ii. (Element 1a) the method comprising initiating the administration of colchicine at a daily low dose to the patient

Petitioner contends that Nidorf teaches "administering colchicine as secondary preventative therapy for 'patients with objectively diagnosed and clinically stable coronary disease,' including patients who had an MI.'" Pet. 36 (citing Ex. 1007, 20:3–5; Ex. 1049 ¶¶ 68, 145–146). Petitioner also asserts that the "COLCOT Protocol also teaches initiating colchicine treatment within 30 days post-MI." *Id.* (citing Ex. 1006, 1 (targeting "patients after myocardial infarction"); Ex. 1049 ¶¶ 78, 146).

Patent Owner opposes. PO Resp. 34–36. Patent Owner contends that Petitioner is focused on one embodiment disclosed in Nidorf and does not consider the prior art as a whole. *Id.* at 35. Patent Owner argues that the LoDoCo trial results disclosed in Nidorf "were of limited value because that study was not placebo controlled." *Id.* (citing Ex. 2001 ¶¶ 21–22). Nidorf "provides a single embodiment wherein 0.5 mg dose is administered with respect to the LoDoCo study but does not describe it to be a preferable amount." *Id.* at 38.

A reference may be relied upon for all that it would have reasonably suggested to one having ordinary skill in the art, including nonpreferred embodiments. *Merck & Co. v. Biocraft Labs., Inc.* 874 F.2d 804 (Fed. Cir. 1989), *cert. denied,* 493 U.S. 975 (1989). Furthermore, "[t]he prior art's mere disclosure of more than one alternative does not constitute a teaching away from . . . alternatives because such disclosure does not criticize, discredit, or otherwise discourage the solution claimed." *In re Fulton*, 391 F.3d 1195, 1201 (Fed. Cir. 2004).

Here, Nidorf teaches using colchicine for use in the treatment and/or prevention of any inflammatory disorder involving the heart tissue. Ex. 1007, 10:1–3. Nidorf teaches administering "a daily dose of colchicine of between about 0.1 mg and about 0.75 mg or between about 0.1 mg and about 0.5 mg." *Id.* at 15:65–16:1. Nidorf concludes "that the addition of colchicine 0.5 mg/day to standard therapy in patients with stable coronary disease significantly reduces the risk of a cardiovascular event, including an ACS, out of hospital cardiac arrest and non-cardio-embolic ischemic stroke." *Id.* at 21:55–59. Based on these disclosures in Nidorf, we find that that Petitioner as demonstrated by a preponderance of evidence that element 1a is taught in Nidorf.

We are not persuaded by Patent Owner's contention that the disclosures in Nidorf are of limited value because they are not placebo controlled. Just because improvements can be made to an experimental protocol does not mean that any information that can be gleaned from even an imperfect experiment should be ignored as Patent Owner urges us to do. "[T]he issue of obviousness is not determined by what the references expressly state but by what they would reasonably suggest to one of ordinary skill in the art." *In re DeLisle*, 406 F.2d 1386, 1389 (CCPA 1969) (citing *In re Siebentritt*, 372 F.2d 566 (CCPA 1967)).

In this case, Petitioner does not solely rely Nidorf to arrive at a method of initiating low dose colchicine treatment in a patient (element 1a). Petitioner also cites the COLCOT protocol for administering colchicine treatment to patients. Pet. 37 (citing Ex. 1006, 2; Ex. 1049 ¶ 1049). The COLCOT protocol describes treating "[p]atients who have suffered a documented acute myocardial infarction within the last 30 days" with colchicine at a dose of 0.5 mg/day. Ex. 1006, 1.

Based upon our review of the parties' arguments and supporting evidence, we determine that Petitioner as demonstrated by a preponderance of evidence that element 1a would have been suggested by the combination of Nidorf and the COLCOT protocol.

iii. (Element 1b) wherein the colchicine is administered without pre-loading the patient with a higher dose of colchicine before the administration at the daily low dose

Petitioner contends that the prior art teaches low dose colchicine administration without a pre-loading dose. Specifically, Petitioner contends that "Nidorf's disclosure that LoDoCo administered 0.5 mg/daily colchicine without a preloading dose" and that the COLCOT protocol "similarly discloses administering 0.5 mg/daily colchicine without a preloading dose." Pet. 37 (citing Ex. 1007, 18:18–24; Ex. 1006, 2; Ex. 1049 ¶¶ 143, 146).

Patent Owner argues that "Nidorf teaches away from the 'no pre-loading dose' limitation." PO Resp. 35. Arguing that a person of ordinary skill in the art "reading Nidorf at the time of the invention would have understood that it disclosed investigating the effect of adding 0.5 mg colchicine to standard therapies, which at the time would include initiating treatment with a loading dose of colchicine." *Id.* at 36 (emphasis omitted). Patent Owner argues that "Nidorf provides a laundry list of acceptable doses that range from 0.1 mg and 5.0 mg. . . . Nowhere does Nidorf teach that the use of 0.5 mg dose is *preferable*. It provides a single embodiment wherein 0.5 mg dose is administered with respect to the LoDoCo study but does not describe it to be a preferable amount." *Id.* at 38 (citing Ex. 1007, 16:55–17:6); Sur-reply 17. Patent Owner's argues that Nidorf should be interpreted as necessarily including a pre-loading dose. *See* PO Resp. 36 ("A POSA reading Nidorf at the time of the invention would have understood that it disclosed investigating the effect of adding 0.5 mg colchicine to ***standard therapies***, which at the time would include initiating treatment with a loading dose of colchicine.").

We disagree with Patent Owner, a reasonable reading of Nidorf's disclosure is that the reference contemplates administering colchicine *with and without a preloading dose*. Nidorf recites:

> In a preferred embodiment, the amount/concentration of colchicine as used herein can be administered at the first day of administration in a higher dose (concentration/amount) compared to the administration of colchicine at the following days(s) of administration (maintenance administration/ maintenance dose of administration). . . . Preferably, the composition of the invention is administered with a dose of colchicine of between about 1.0 mg to about 2.0 mg at the first day (preferably as a single dose) of administration and the maintenance dose of colchicine at the following day(s) of administration is between about 0.5 mg to about 1.0 mg.

Ex. 1007, 16:8–28.

"A reference does not teach away . . . if it merely expresses a general preference for an alternative invention but does not 'criticize, discredit, or otherwise discourage' investigation into the invention claimed." *DePuy Spine, Inc. v. Medtronic Sofamor Danek, Inc.*, 567 F.3d 1314, 1327 (Fed. Cir. 2009) (citing *In re Fulton*, 391 F.3d 1195, 1201 (Fed. Cir. 2004)). Here, Nidorf may have indicated a preference for a preloading dose, but Nidorf's disclosure also encompasses administering colchicine *with and without* a preloading dose to patients at risk of cardiovascular event.

Nidorf teaches that "[i]n an embodiment, the composition according to the present invention is administered at a daily dose of about 0.5 mg colchicine." Ex. 1007, 16:5–7. "Patients randomized to active treatment [group in Nidorf] were given a prescription for colchicine 0.5 mg daily by their referring cardiologist. The drug was dispensed by their usual chemist, and if requested, patients were reimbursed for the cost of these scripts. All other treatments were continued as usual." *Id.* at 18:65–19:3. Here, there is no disclosure in Nidorf of telling the patient to take more colchicine on the first day.

Additionally, claims 1–3 of Nidorf read as follows:

> 1. A method of treating and/or reducing the risk of a cardiovascular event in a subject, the method comprising:
>
> > administering colchicine, a salt thereof, and/or any combination thereof to a subject at risk of a cardiovascular event or who has had a cardiovascular event;
> >
> > wherein the cardiovascular event is acute coronary syndrome, out-of-hospital cardiac arrest, or noncardioembolic ischemic stroke.
>
> 2. The method of claim 1, wherein the composition is administered once per day.
>
> 3. The method claim 1, wherein about 0.5 to about 0.75 mg of colchicine, a salt thereof, and/or any combination thereof is administered to the subject.

Ex. 1007, 22:12–24.

Here, Nidorf's claims do not specifically recite using a pre-loading but the claims are written broadly enough that they would encompass administration of colchicine *with and without a preloading dose.*

Accordingly, we are not persuaded by Patent Owner's argument that Nidorf should be interpreted as *necessarily* including a pre-loading dose with the colchicine administration. *See* PO Resp. 36. Patent Owner argues that "[t]he prior art also corroborates the understanding that a loading dose should be used." *Id.* at 39 (citing Ex. 2001 ¶¶ 18, 25, 28, 36–40, 44, 56; Ex. 1040; Ex. 2007; Ex. 2005; EX2006; Ex. 2002; Ex. 1020). Just because Patent Owner's expert can direct us to references that utilized a pre-loading dose in their colchicine administration protocols does not persuade us that other clinical protocols that do not recite such a pre-loading should be interpreted as including such a dose. *See* Ex. 1006; Ex. 1007, 22:12–24, Ex. 1020, 11 ("patients with established coronary disease presenting for routine clinical review were randomized to receive colchicine 0.5 mg/day or no colchicine without any other changes to their medical therapy");Ex. 1075 ¶ 49 ("A POSA would understand that. . . [b]ecause dosing errors are so commonly made and can be so dangerous, physicians are trained to be explicit when they talk about dose requirements to their patients as well as in all written documents, including clinical trial protocols and results").

Furthermore, Petitioner's unpatentability challenge is not premised on Nidorf alone but relies on a combination of references. We have reviewed the Nidorf and the COLCOT protocol and agree with Petitioner that these protocols include methods that *do not* recite administering a pre-loading dose to patients. *See* Pet. 37 (citing Ex. 1007, 18:18–24; Ex. 1006, 2; Ex. 1049 ¶¶ 143, 146); Ex. 1049 ¶¶ 44–56, 67. This is supported by Petitioner's expert, Dr. Ridker, describing that Nidorf and the COLCOT protocol administer 0.5 mg/day to a patient with no mention to include a pre-loading dose in these protocols. Ex. 1049 ¶¶ 67 ("Nidorf describes the 'LoDoCo' trial that 'assess[ed] the therapeutic effects" of colchicine in 'patients with cardiovascular disease' by administering 0.5 mg/day of colchicine without a preloading dose to patients with clinically stable coronary disease.")(footnote omitted), 79 ("COLCOT Protocol describes administering a 0.5 mg tablet of colchicine once per day to the patients.") (footnote omitted), 146.

Based upon our review of the parties' arguments and supporting evidence in the full record, we determine that Petitioner has demonstrated by a preponderance of the evidence that the combination of Nidorf and COLCOT protocol teach the administration of colchicine without a pre-loading dose as recited in element 1b.

iv. (Element 1c) within about 3 days of the MI

Petitioner's expert acknowledges that Nidorf does not teach initiating the treatment "within about 3 days" of the myocardial infraction. Ex. 1049 ¶ 145. Petitioner's expert finds that the COLCOT protocol provides a 30-day window in which to start the colchicine therapy, and this would include starting therapy within 3 days of the myocardial infarction. *Id.* Petitioner contends that "[i]nitiation times within this prior art range are presumptively obvious and not overcome here by any unexpected results or evidence of criticality. *E.I. DuPont de Nemours & Co. v. Synvina C.V.*, 904 F.3d 996, 1006 (Fed. Cir. 2018) ("presumption of obviousness" for values within prior art range)." Pet. 38–39. Petitioner contends that this is further supported by the ESC guidelines teaching "initiating secondary preventative therapy at or prior to post-MI hospitalization discharge, corresponding to around day 2 or 3 post-MI." Pet. 39 (citing Ex. 1009, 24, 30, 33; Ex. 1049 ¶ 159) (footnote omitted). Petitioner contends "that initiating treatment in stabilized post-MI patient at or before hospitalization discharge would potentially provide increased compliance and earlier prevention of secondary cardiovascular events." Pet. 42 (citing Ex. 1009, 33; Ex. 1008, 6; Ex. 1049 ¶ 171).

Patent Owner argues that Petitioner did not explain why a person of ordinary skill in the art at the time the invention was made "would have been motivated to shorten Nidorf's [time to treatment initiation] TTI requiring the patient to be stable for at least 6-months after an MI" to arrive at the claimed time frame of "within about 3 days of the MI" as recited in element 1c. PO Resp. 41; Sur-Reply 19. Patent Owner argues that there is no motivation "to shorten Nidorf's treatment initiation time to around 2-3 days post-MI." PO Resp. 41. Patent Owner argues that Petitioner presented only conclusory arguments with respect to the start of colchicine treatment and provides no evidence why early treatment, such as 3 days after a myocardial infarction would have been effective. *See id.* at 41 (citing *KSR*, 550 U.S. at 418 (quoting *In re Kahn*, 441 F.3d 977, 988 (Fed. Cir. 2006)). In other words, Patent Owner contends that Petitioner has not articulated (a) a motivation for arriving at the early time to treatment; (b) has not shown a reasonable expectation of success; and (c) has not shown that based on routine optimization one of ordinary skill in the art would arrive at the claimed limitation. We address Patent Owner's contentions in turn below.

a) Motivation

We have reviewed the parties' contentions and find that Petitioner has provided a sufficiently articulated reason for arriving at the limitation of element 1c "within about 3 days of the MI" as claimed. Specifically, Petitioner cites improved patient compliance as a motivating factor for starting therapy before patients leave the hospital after suffering an MI. *See* Pet. 41 (citing Ex. 1009, 33; Ex. 1049 ¶¶158-165); Reply 17–18.

Patent Owner's argument that there is no reason to shorten Nidorf's time to treatment initiation (TTI) focuses on a single reference. PO Resp. 41; Reply 19. Petitioner's unpatentability ground, however, relies on the combination of four references and does not solely rely on the teaching of Nidorf to arrive at the "within about 3 days of the MI" limitation. *See* Pet. 38–41. "Non-obviousness cannot be established by attacking references individually where the rejection is based upon the teachings of a combination of references. . . . [The reference] must be read, not in isolation, but for what it fairly teaches in combination with the prior art as a whole." *In re Merck & Co*., 800 F.2d 1091, 1097 (Fed. Cir. 1986).

Petitioner acknowledged that that Nidorf does not teach initiating the treatment "within about 3 days" of the myocardial infraction but finds that the COLCOT protocol suggest starting colchicine treatment within a 30-day window of the myocardial infarction. Pet. 38–39; Ex. 1049 ¶¶ 145, 157, 159. Petitioner further relies on the teaching in the ESC Guidelines to suggest that therapy should preferably be started on or prior to post-MI hospitalization discharge. Pet. 39.

Petitioner contends that the average post-MI hospital stay for low risk patients is around 2 or 3 days. Pet. 39 (citing Ex. 1049 ¶ 157; Ex. 1009, 33, Ex. 1011, 1 ("early discharge from the hospital at day 2 or sooner of these low-risk patients does not appear to be associated with an increased risk of adverse events post discharge compared with those discharged at a later time"); Ex. 1014, Abstract ("Early identification of low risk patients with MI allowed safe omission of the intensive care phase and noninvasive testing, and a day 3 hospital discharge strategy, resulting in substantial cost savings").

This is further supported by the ESC guidelines that describe the length of hospital stay after ST-segment elevation myocardial infarction (STMI) to be relatively short.

> Several studies have shown that low-risk patients with successful primary [percutaneous coronary intervention] PCI and complete revascularization can safely be discharged from hospital on day 2 or day 3 after PCI. . . . A short hospital stay implies limited time for proper patient education and up-titration of secondary prevention treatments. Consequently, these patients should have early post-discharge consultations with a cardiologist, primary care physician, or specialized nurse scheduled and be rapidly enrolled in a formal rehabilitation programme, either in-hospital or on an outpatient basis.

Ex. 1009, 24; Pet. 14–15. "Low treatment adherence is an important barrier to achieving optimal treatment targets and is associated with worse outcomes.[] Delayed outpatient follow-up after [acute myocardial infarction] AMI results in worse short- and long-term medication adherence." Ex. 1009, 30. "Lipid-lowering treatment should be started as early as possible, as this increases patient adherence after discharge, and given as high-intensity treatment, as this is associated with early and sustained clinical benefits." Ex. 1009, 33.

We agree with Petitioner that the COLCOT protocol suggests treating patients within 30 days of the myocardial infarction, and this would reasonably encompass treating patients within 3 days of the event. *See* Pet. 38–41; *see* Ex. 1006, 1 ("Patients who have suffered a documented acute myocardial infarction within the last 30 days, are treated according to the national guidelines and after having completed any planned percutaneous revascularization procedures associated with their initial infarction will receive either colchicine (0.5 mg per day).”). COLCOT protocol inclusion criteria requires that the patient have a documented myocardial infarction and treatment needs to start before 30 days has elapsed. *See* Ex. 1006, 1. The range for administration of colchicine in the COLCOT protocol, therefore, is reasonably interpreted to be from day 0 through day 30. *Id.*; Ex. 1049 ¶ 150.

Petitioner further supports the position for the early administration of secondary preventative therapies with the testimony by Dr. Ridker. *See* Pet. 39–40; Ex. 1049 ¶¶ 147–153. Petitioner’s expert avers that “[i]n my experience, a 3-day post-MI hospitalization stay was common in 2019 as it is today. Consistent with common medical practice . . . that pre-discharge management included the administration of secondary preventive therapy (in particular statins, which had been approved for this indication) to ‘increase patient adherence after discharge’ and provide ‘early and sustained clinical benefits.’” Ex. 1049 ¶ 148 (citing Ex. 1009, 24 (“A short hospital stay implies limited time for proper patient education and up-titration of secondary prevention treatments.”), 33 (“Therefore, statins are recommended in all patients with AMI . . . Lipid-lowering treatment should be started as early as possible, as this increases patient adherence after discharge, and given as high-intensity treatment, as this is associated with early and sustained clinical benefits.”) (footnote omitted). Petitioner’s expert Dr. Ridker further avers that before 2019, multiple other clinical trials initiating low dose colchicine in MI patients early post-MI had been already registered, initiated, or completed. Ex. 1049 ¶ 49.

For the reasons discussed above, we determine that Petitioner has demonstrated by a preponderance of evidence that there was a sufficiently articulated motivation for reducing the time to treatment initiation colchicine administration for the patient to coincide with the discharge from the hospital after suffering from a myocardial infarction.

b) Reasonable Expectation of Success

Petitioner contends that Nidorf had already successfully demonstrated reducing "'the risk of a cardiovascular event, including an ACS, out of hospital cardiac arrest and non-cardio-embolic ischemic stroke' in cardiovascular disease patients (including those with past acute MI) that had been stable for at least 6 months by administering 0.5 mg daily colchicine." Pet. 42 (citing Ex. 1007, 21:55–59, 20:34–36; Ex. 1049 ¶¶ 28, 69, 167). Petitioner contends that "[b]uilding on Nidorf's demonstrated efficacy, a POSA would have expected that initiating colchicine treatment at about 3 days post-MI would also be effective to reduce the risk of future cardiovascular events." *Id.* (citing Ex. 1049 ¶ 151). Petitioner contends that by 2019 one of ordinary skill in the art would have been aware of the "inflammation hypothesis" of cardiovascular disease proven by CANTOS in patients at least 30 days post-MI. Pet. 43 (citing Ex. 1039, 19; Ex. 1044, 3; Ex. 1026, 4, Figure 1; Ex. 1049, ¶¶ 47, 168–171). And further reducing the time to initiate treatment to less than 30 days (i.e. a range from 0–30) as taught by the COLCOT protocol would reasonably have expected similar success. Pet. 43 (citing Ex. 1009, 33; Ex. 1006, 2; Ex. 1049 ¶ 171).

The question of obviousness cannot be approached on the basis that an artisan having ordinary skill would have known only what was read in the references, because such artisan must be presumed to know something about the art apart from what the references disclose. *See In re Jacoby*, 309 F.2d

513, 516 (CCPA 1962). Dr. Ridker explains that well before 2019, the CANTOS trial provided "definitive evidence that directly targeting inflammation is beneficial for the secondary prevention of cardiovascular disease." Ex. 1049 ¶ 168; Pet. 42; Ex. 1032–1035. "CANTOS involved the administration of another anti-inflammatory agent (canakinumab)" and "that its proof of the inflammatory hypothesis was applicable to other agents" that are active in innate immunity. Ex. 1049 ¶ 169; Ex. 1032–1035.

According to Petitioner, "CANTOS was widely recognized as providing significant clinical validation that targeting inflammation—as colchicine does—reduces cardiovascular risks." Pet. 43 (citing Ex. 1033, 1; Ex. 1032, 1; Ex. 1034, 1 ("The CANTOS trial . . . now provides the first largescale proof of concept that inflammasome targeting can reduce cardiovascular events"); Ex. 1035, 1; Ex. 1015, 1 ("the CANTOS trial showed that directly reducing inflammation with canakinumab, an interleukin (IL)-1β neutralizing monoclonal antibody, could also reduce cardiovascular event rates"); Ex. 1026, 4, Figure 1; Ex. 1047, 6; Ex. 1049 ¶¶ 47, 169). Petitioner further explains that Deftereos also teaches "safely improving biomarkers by colchicine initiation with 12 hours post-MI." Pet. 43 (citing Ex. 1049 ¶ 171; Ex. 1019, 3; Ex. 1006, 1; Ex. 1040, 1 ("Patients presenting with ST-segment–elevation myocardial infarction ≤12 hours from pain onset . . . were randomly assigned to colchicine or placebo for 5 days. . . . [and] results suggest a potential benefit of colchicine in ST-segment–elevation myocardial infarction")). Petitioner's position is that before 2019 the art already suggests targeting inflammation in order to reduce cardiovascular risks. Petitioner contends that "a POSA would have been motivated to initiate administration of colchicine—already shown effective in LoDoCo—prior to or at hospitalization discharge, i.e., at about 2-3 days of the MI, to 'increase[] patient adherence' and to obtain 'early and sustained clinical benefits.'" Pet. 41 (citing Ex. 1009, 33; Ex. 1049 ¶¶ 158-165).

"Obviousness does not require absolute predictability of success . . . all that is required is a reasonable expectation of success." *In re Droge*, 695 F.3d 1334, 1338 (Fed. Cir. 2012) (quoting *In re Kubin*, 561 F.3d 1351, 1360 (Fed. Cir. 2009) (citing *In re O'Farrell*, 853 F.2d 894, 903–04 (Fed.Cir.1988)); *Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359 (Fed. Cir. 2016) (explaining that the expectation of success issue involves a showing of "a reasonable expectation of achieving *what is claimed*") (emphasis added). As Petitioner explains, Nidorf already successfully demonstrated reducing "the risk of a cardiovascular event, including an ACS, out of hospital cardiac arrest and non-cardio-embolic ischemic stroke" in cardiovascular disease patients (including those with past acute MI) that had been stable for at least 6 months by administering 0.5 mg daily colchicine. Pet. 42 (citing Ex. 1007, 21:55–59, 20:34–36; Ex. 1049 ¶¶ 28, 69, 167). The COLCOT protocol suggests administering colchicine to stable patients that had a myocardial infarction within the last 30 days. Ex. 1006, 1.

We also agree with Petitioner that based on the combination of Nidorf and the COLCOT protocol a person of ordinary skill in the art at the time the invention was made would have been motivated to administer colchicine anywhere between 0–30 days as suggested by the COLCOT protocol. Here, as Petitioner explains the person of ordinary skill in the art would have appreciated that it was possible to improve levels of "biomarkers by colchicine initiation with 12 hours post-MI." Pet. 43 (citing Ex. 1049 ¶ 171; Ex. 1019, 3; Ex. 1006, 1; Ex. 1040, 1); Ex. 1049 ¶ 151. Thus, taken together we agree with Petitioner that based on what was known in the art at the time the invention was made there was a reasonable expectation that administering colchicine early would similarly result in reducing inflammation which would be beneficial for the secondary prevention of cardiovascular disease.

We have reviewed the parties' contentions but are not persuaded by Patent Owner's argument that the art is unpredictable. Specifically, Patent Owner contends that "[o]ther studies investigated use of low-dose colchicine in the more acute settings, and none of which showed any improvement as compared to placebo." PO Resp. 57 (citing Ex. 2072 ¶¶ 12–17, 34–36). Patent Owner contends that prior art "showed that initiation of low-dose colchicine in the acute setting was not effective for treating or preventing CV events." PO Resp. 58. Dr. Ardehali, Patent Owner's expert cites various studies to show that "the effectiveness of colchicine in treating or preventing CV events was highly unpredictable. Ex. 2072 ¶ 16 (citing Ex. 2009, 4; Ex. 2061; Ex. 2054; Ex. 2064; Ex. 1020; Ex. 1028; Ex. 2055). Specifically, Dr. Ardehali avers that "more than thirty years ago O'Keefe found that colchicine was not effective compared to placebo at preventing CV disease in the studied population." Ex. 2072 ¶ 12 (citing Ex. 2009). According to Dr. Ardehali

> the body of research has shown that using colchicine to target the inflammatory pathway is not consistently effective. *See, e.g.*, EX1028 (concluding low-dose colchicine "was not associated with a significantly increased likelihood of achieving a CRP level <2 mg/L or

> lower absolute levels of CRP 30 days after an acute MI"); EX2054 ("[O]ur pilot study provided no evidence that colchicine 1 mg daily for 30 days compared with placebo suppresses inflammation in patients with acute coronary syndrome or acute ischemic stroke."); EX2055 ("In our study, the effect of colchicine on inflammation in the context of STEMI could not be demonstrated."); EX2056 ("The addition of colchicine to standard medical therapy did not significantly affect cardiovascular outcomes at 12 months in patients with ACS and was associated with a high rate of mortality.").

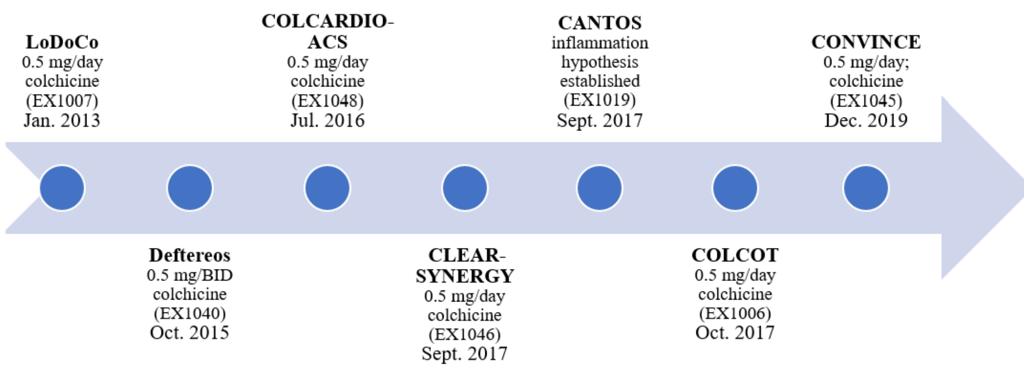
Ex. 2072 ¶ 12. Therefore, Patent Owner argues that a person of ordinary skill in the art at the time the invention was made "would have been skeptical that initiating low-dose colchicine, without a pre-loading dose, to patients within 3 days of an MI would reduce the risk of any CV events, including stroke." PO Resp. 58–59 (citing Ex. 2072 ¶ 37); Sur-reply 23.

In response, Dr. Ridker explains that Patent Owner's references can be sorted into two groups. Ex. 1075 ¶¶ 8–9 (emphasis added), 14–15 (citing PO Resp. 5):

> Group 1: low-dose colchicine was *not effective* to treat certain CV events when compared to placebo. These studies include O'Keffe (Ex. 2009); Raju (Ex. 2054), Hennessy (Ex.1028), Akodad (Ex. 2055), and Tong (Ex. 2056).
>
> Group 2: low-dose colchicine [in stable patients] could be *effective* at treating certain CV events when compared to placebo. These studies include Deftereos 2013 (Ex. 2064), Nidorf 2013 (Ex. 1020), Nidorf 2020 (Ex. 2065), and Tardif 2019 (Ex. 1022).

As Dr. Ridker explains the studies O'Keffe (Ex. 2009); Raju (Ex. 2054), Hennessy (Ex.1028), Akodad (Ex. 2055), and Tong (Ex. 2056). contained some flaws in that it relied on a small sample size and that it used "surrogate endpoints as primary outcome (such as change in hsCRP, flow-mediated dilation, or troponin I measurement) that may or may not ultimately prove informative when large hard outcomes trials are conducted." Ex. 1075 ¶ 19. Dr. Ridker further explains that surrogate end points can be considered informative but given the more recent studies showing "the effect of low-dose colchicine on hard clinical endpoints was already established by LoDoCo and LoDoCo2." *Id.*

A time line presented by Dr. Ridker showing the various studies administering anti-inflammatory compositions post-MI is reproduced below:



The figure shows a time-line for various studies using anti-inflammatory agents after an MI-event. *See* Ex. 1049 ¶ 161. Dr. Ridker explains that well before 2019, the CANTOS trial provided "definitive evidence that directly targeting inflammation is beneficial for the secondary prevention of cardiovascular disease." Ex. 1049 ¶ 168; Pet. 42; Ex. 1032–1035. "CANTOS involved the administration of another anti-inflammatory agent (canakinumab)" and "that its proof of the inflammatory hypothesis was applicable to other agents" that are active in innate immunity. Ex. 1049 ¶ 169; Ex. 1032–1035.

Three large scale randomized placebo controlled clinical trials Nidorf 2013 (Ex. 1020 (LoDoCo)), Nidorf 2020 (Ex. 2065 (LoDoCo2)), and Tardif 2019 (Ex. 1022 (COLCOT)) also show that a dose of 0.5 mg once daily had already been established as a highly effective method to reduce cardiovascular event rates in patients with or at high risk for chronic atherosclerotic disease. Ex. 1075 ¶ 6 (citing Ex. 1051–1053, Ex. 1074).

As explained above, obviousness does not require absolute predictability of success all that is required is a "a reasonable expectation of achieving what is claimed." *Intelligent Bio-Systems*, 821 F.3d 1359. Here, Petitioner has demonstrated by a preponderance of evidence that based on the knowledge of one of ordinary skill in the art at the time the invention was made that there was an expectation that suppressing inflammation in a patient suffering a recent MI would reduce future cardiovascular event rates.

c) Routine Optimization

Petitioner contends, and we agree, that reducing the 30-day window as taught in the COLCOT protocol would have been a matter of routine optimization. Reply 16; Pet. 38 ("the claimed 'within about 3 days of the MI' would have been a matter of routine optimization within COLCOT Protocol's 30-day window. . . . Initiation times within this prior art range are presumptively obvious").

Patent Owner contends that "a *prima facie* case [based on overlapping ranges] is not presumed because Petitioner has not met its burden of production that the COLCOT protocol is prior art." PO Resp. 42.

For the reason discussed above (II.C.2.a.i), based on the totality of the evidence provided by Petitioner we are persuaded that Exhibit 1006 was publicly accessible more than one year prior to filing date of Nov. 15, 2019. Reply 10–11. Exhibit 1006 appears to be identical to the document retrieved by Dr. Mullins using the same URL. *Compare* Ex. 1006 *with* Ex. 1073, 31–35 (attachment A-1). In addition, Dr. Mullins directed us to two contemporaneous articles that identify the same COLCOT protocol associated with the same URL. *See* Ex. 1073 ¶¶ 23–28 (attachment A–2 and A–3). Based on these disclosures, we determine that the totality of the evidence corroborates the testimony of Mr. Cruitt (Ex. 1038; Ex. 1072) or Dr. Mullins (Ex. 1073), or both, and is sufficient to show that Exhibit 1006 was publicly accessible more than one year prior to the filing date of the ’063 patent. Accordingly, we are not persuaded by Patent Owner’s contention that the COLCOT protocol is not prior art.

Patent Owner contends that discovering the benefit of early colchicine treatment was “not a matter of routinely optimizing COLCOT’s 30-day window.” PO Resp. 41–42 (citing *Novartis Pharmas. Corp. v. West-Ward Pharmas. Int’l Ltd.*, 923 F.3d 1051 (Fed. Cir. 2019).

We agree with Petitioner that *Novartis* neither addresses routine optimization nor limits its discussion to composition claims. Reply 17.

> In the context of claimed numerical ranges, . . . we have explained that an overlap between a claimed range and a prior art range creates a presumption of obviousness that can be rebutted with evidence that the given parameter was *not* recognized as result-effective. *See Genentech, Inc. v. Hospira, Inc.*, 946 F.3d 1333, 1341 (Fed. Cir. 2020) (citing *E.I. DuPont*, 904 F.3d at 1006); [] *In re Applied Materials, Inc.*, 692 F.3d 1289, 1295 (Fed. Cir. 2012).

Pfizer Inc. v. Sanofi Pasteur Inc., 94 F.4th 1341, 1347-1348 (Fed. Cir. 2024).

Here, Petitioner is relying on the COLCOT protocol to teach the initiation of colchicine administration to patient populations having suffered a myocardial infarction within the last 30 days. Pet. 38 (citing Ex. 1006, 1; Ex. 1049 ¶ 147). We interpret "within the last 30 days" as recited in the COLCOT protocol to encompass initiation of colchicine to a patient in a range from 0–30 days after suffering from a myocardial infarction.

We agree with Petitioner that this is a case of routine optimization. Reply 16 (citing *Pfizer Inc. v. Sanofi Pasteur Inc.*, 94 F.4th 1341, 1347 (Fed. Cir. 2024)1347). *Pfizer* explains that "[a] routine optimization analysis generally requires consideration whether a person of ordinary skill in the art would have been motivated, with a reasonable expectation of success, to bridge any gaps in the prior art to arrive at a claimed invention." *Pfizer*, 94 F.4th at 1347–1348. For the reason discussed above (II.D.4.b.ii–iii), we determine that Petitioner has sufficiently articulated a motivation that to improve patient compliance it is important to start administration of colchicine as soon as the patient is stable and preferably before they leave the hospital. We also find that there is a reasonable expectation of success in the art at the time the invention was made that administering colchicine early after a myocardial infarction at least reduces inflammation markers.

d) Element 1c Summary

Based upon our review of the parties' arguments and supporting evidence in the full record, we determine that Petitioner has demonstrated by a preponderance of the evidence that the combination of Nidorf and COLCOT protocol teach the administration of colchicine within 0–30 days of the myocardial infarction, thereby providing a presumption of obviousness for claim element 1c because the values of administrating with 0–3 days falls within the prior art range.

v. (Element 1d) wherein percutaneous coronary intervention was performed for treating the patient's MI.

Petitioner contends that a patient population including those that had received percutaneous coronary intervention (PCI) is taught by COLCOT protocol. Pet. 37 (citing Ex. 1006, 2, 5; Ex. 1049 ¶¶ 23, 78).

Here, COLCOT protocol discloses that the relevant patient population includes patients that had received percutaneous coronary intervention (PCI) therapy. COLCOT protocol describes the patient population as "[p]atients who have suffered a documented acute myocardial infarction within the last 30 days, are treated according to the national guidelines *and* after having completed any planned percutaneous revascularization procedures associated with their initial infarction will receive . . . colchicine (0.5 mg per day) . . . for an estimated 2 year[] period." Ex. 1006, 1 (emphasis added), 4 (emphasis added) ("Patient must have completed *any* planned percutaneous revascularization procedures associated with his or her qualifying myocardial infarction"). COLCOT protocol, therefore, discloses a genus of patients having experienced a myocardial infarction, and further identifies a subgroup (species) of patients that have had myocardial infarction *and* have received percutaneous coronary intervention (PCI) therapy. COLCOT protocol discloses that both populations are treated the same way – they are administered 0.5 mg/day of colchicine starting within 30-days of the myocardial infarction.

In *In re Petering*, our reviewing court found that in some circumstances species claims were unpatentable over prior art that was directed to a genus:

> It is our opinion that one skilled in this art would, on reading the [prior] patent, *at once envisage each member of this limited class,* even though this skilled person might not at once define in his mind the formal boundaries of the class as we have done here . . . [W]e think that one with ordinary skill in this art, with . . . [the reference genus patent] before him, *would also have before him those subspecies.*

In re Petering, 301 F.2d 676, 681–82 (1962) (emphases added). "[T]he question under 35 USC 103 is not merely what the references expressly teach but what they would have suggested to one of ordinary skill in the art at the time the invention was made." *Merck & Co. v. Biocraft Laboratories Inc.,* 874 F.2d 804, 807 (Fed. Cir. 1989).

Here, the COLCOT protocol teaches treating a patient having had a myocardial infarction by administering 0.5 mg/day of colchicine starting within 30-days of the myocardial infarction, and it does not treat patients that received percutaneous coronary intervention (PCI) therapy differently from those that did not get PCI treatment. The method disclosed in the COLCOT protocol, therefore, administers colchicine the same way regardless of whether the patient did or did not undergo a percutaneous revascularization procedure. Thus, the COLCOT protocol encompasses treating patients that received percutaneous coronary intervention after suffering a MI and thereby meets the limitation of element 1d.

vi. Objective Indicia of Nonobviousness

A *prima facie* case of obviousness may be rebutted by providing a "showing of facts supporting the opposite conclusion." Such a showing dissipates the *prima facie* holding and requires us to "consider all of the evidence anew." *In re Kumar*, 418 F.3d 1361, 1368 (Fed. Cir. 2005) (citations omitted). One way to "rebut a *prima facie* case of obviousness is to make a showing of 'unexpected results,' *i.e.,* to show that the claimed invention exhibits some superior property or advantage that person of ordinary skill in the relevant art would have found surprising or unexpected." *In re Soni*, 54 F.3d 746, 750 (Fed. Cir. 1995). "[W]hen unexpected results are used as evidence of nonobviousness, the results must be shown to be unexpected compared with the closest prior art." *In re Baxter Travenol Labs.,* 952 F.2d 388, 392 (Fed. Cir. 1991). "After a *prima facie* case of obviousness has been made and rebuttal evidence submitted, all the evidence must be considered anew." *In re Eli Lilly & Co.,* 902 F.2d 943, 945 (Fed. Cir. 1990) (citing *In re Piasecki,* 745 F.2d 1468, 1472 (Fed. Cir. 1984)).

Patent Owner contends that "[t]he ability to treat CV events and prevent future CV events with the claimed treatment regimen is an unexpected result." PO Resp. 55. Patent Owner acknowledges that the LoDoCo study showed a benefit in low-dose colchicine treatment, however, the treatment was not initiated until six months after the patient is stabilized. PO Resp. 57 (citing Ex. 1007, Ex. 1020). Patent Owner contends that the risk reduction of 23% for ischemic CV events in post-MI patients when colchicine treatment was initiated at less than 3 days compared with placebo is unexpected. PO Resp. 57. According to Dr. Ardehali, Patent Owner's expert, "Dr. Tardif's metanalysis of the data from four randomized controlled trials, surprisingly showed there was a 62% attenuated risk reduction in stroke for patients administered low-dose colchicine according to claims 27-39." Ex. 2001 ¶ 59.

Petitioner disagrees. Petitioner contends that Patent Owner has not shown criticality of the limitation within about 3 days of the myocardial infarction. Pet. 56. According to Petitioner, the "COLCOT Protocol specifically taught the dominating range of within 30 days post-MI, yet the patent data fail to show any criticality, much less 'unexpected' benefit, of the "within about 3 days of the MI" limitation, compared to later initiation times within disclosed the 30-day range." *Id.* Petitioner contends that "the data in the '063 patent . . . fail to show any significance differences between the TTI groups of 0-3 days, 4-7 days, and 8-30 days." *Id.* (citing Ex. 1001, Table 3; Ex. 1049 ¶¶ 227–228, 234–241; Ex. 1012, 3–4 ("The authors of [Bouabdallaoui] did not report whether the differences in observed effect size between strata were statistically significant."); Ex. 1038 ¶ 29). Dr. Ridker explains that "TTI data [in the '063 patent] is a *post hoc* analysis based on later identified (i.e., not pre-specified) patient COLCOT subgroups. What this means is that the newly defined subgroups may not be sufficiently randomized, which undermines between-group comparisons and conclusions as to relative superiority." Ex. 1049 ¶ 235 (citing Ex. 2021, 6).

"[W]hen unexpected results are used as evidence of nonobviousness, the results must be shown to be unexpected compared with the closest prior art." *In re Baxter Travenol Labs.,* 952 F.2d 388, 392 (Fed. Cir. 1991). The unexpected results must be "different in kind and not merely in degree from the results of the prior art." *In re Applied Materials, Inc.,* 692 F.3d 1289, 1297 (Fed. Cir. 2012). In this case the closest prior art with data is COLCOT (Ex. 1006). "Moreover, the applicant's showing of unexpected results must be commensurate in scope with the claimed range." *In re Peterson*, 315 F.3d 1325, 1330 (Fed. Cir. 2003).

Figure 2 of the '063 patent[10] is reproduced below.

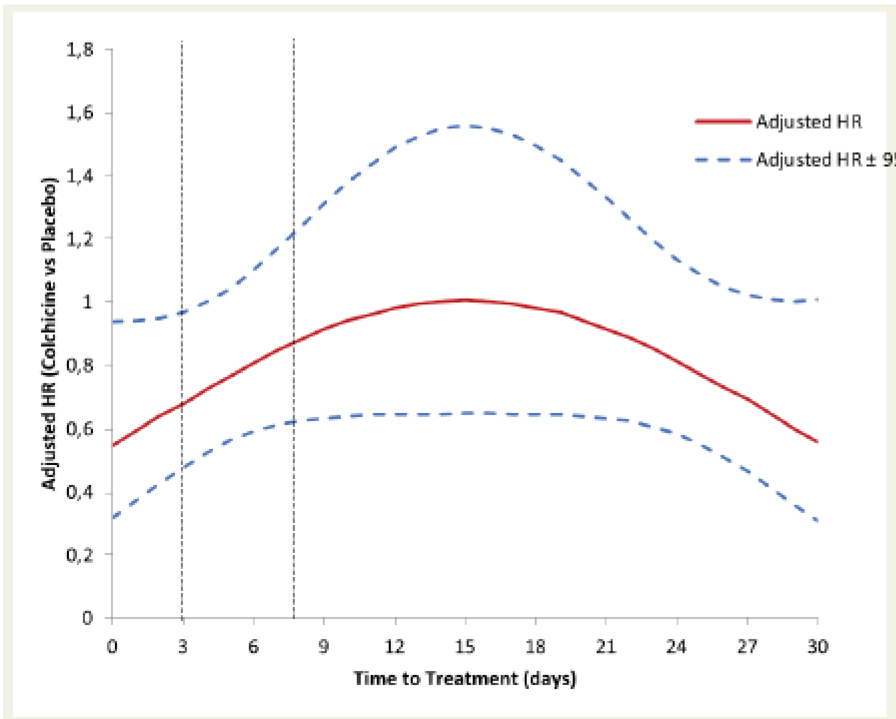


Figure 2 of the '063 patent shown above "is a graph of the association between the time-to treatment initiation and the adjusted hazard ratio (solid [red] line) with the 95% confidence intervals of the adjusted hazard ratio (dashed [blue] line[s]) calculated using a quadratic multivariable Cox regression model." Ex. 1001, 6:47–51; Reply 27.

As Dr. Ridker explains "the Figure 2 graph shows an apparent parabolic effect for treatments initiated early and late in the overall 30-day

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[10] For legibility we use Figure 2 reproduced from Petitioner's Reply. Reply 27 (citing Ex. 1005). Figure 2 from the Bouabdallaoui publication is the same as Figure 2 of the '063 patent. *Compare* Ex. 1005, Figure 2 *with* Ex. 1001, Figure 2.

window. What this means is that initiating treatment within 0-3 days does not provide any different result or benefit relative to initiating treatment late in the 30-day window, e.g., by around day 27 during the last three days of the window." Ex. 1049 ¶ 242. "[I]nstead of achieving superior adjusted HRs (hazard ratios) over all TTI 4-30 days patients, the HRs of TTI 0-3 days patients are almost at the same level as those for TTI 27-30 days." Ex. 1049 ¶ 243. "[A] skilled cardiologist would interpret this parabolic data as reflecting an artifact of the *post hoc* analysis rather than an actual treatment effect." Ex. 1049 ¶ 245 (citing Ex. 1012).

Dr. Ridker further explains that "my reading of these data and that of the field as a whole indicates that the benefits of low dose colchicine are of similar magnitude and persist over very long treatment periods, irrespective of timing of a prior acute coronary syndrome (ACS). In other words, the patent data does not establish differences (beyond chance) between the TTI subgroups." Ex. 1049 ¶ 225. "[T]he '063 patent does not show any unexpected superiority in reducing stroke risks by initiating treatment within a 30 day window compared to the prior art LoDoCo trial that initiated treatment much later in time." Ex. 1049 ¶ 229. The LoDoCo trial published in 2013, reported a hazard ratio of 0.23 (0.03–2.03) for stroke after initiation of colchicine after 6 months which is fully consistent with hazard ratio for stroke 0.26 (0.10-0.70) found in COLCOT protocol with colchicine initiation between 0–30 days. Ex. 1049 ¶ 229 (citing Ex. 1001 ('063 patent), Table 10.). "[T]he data does not show that initiating treatment within 3 days of the MI achieved unexpectedly enhanced stroke reduction compared to initiating treatment anytime later than 3 days." Ex. 1049 ¶ 230. The '063 "patent Table 3 shows that the primary composite endpoint hazard ratio range for 0-3 days overlaps with the ranges for 4-7 and 8-30 days [time to treatment initiation] TTI." Pet. 57 (citing Ex. 1001, Table 3; Ex. 1049 ¶ 230).

Petitioner contends that "the COLCOT study was designed to determine whether a daily low dose of colchicine initiated within 30 days after an MI was beneficial; it was *not* designed to assess outcomes based on the claimed TTI." Reply 23 (citing Ex. 1077, 55:8–21). Petitioner's expert Dr. Ridker avers that "the reported effect on stroke reduction of 1.6% to 0.4% with a hazard ratio of 0.25 [in COLCOT] is nearly the same as the stroke reduction of 0.7% to 0.2% with a hazard ratio of 0.23 reported in LoDoCo." Ex. 1049 ¶ 221; *compare* Ex. 1001 ('063 patent), Table 10 *with* Ex. 1020 (Nidorf 2013), Table 3. In other words, hazard ratio for starting treatment in a stable patient between 0–30 days is nearly the same as the hazard ratio for starting treatment in a stable patient after 6 months. We credit Dr. Ridker's testimony and agree that any difference is at best a difference in the degree but not a difference in kind and, therefore, does not support a finding of unexpected results.

Additionally, Petitioner contends that "[i]t is well known that subgroup analyses are notoriously unreliable. As a seminal paper says, 'Subgroup analyses in clinical trials [are] fun to look at, but don't believe them!'" Reply 24 (citing Ex. 1067, 1); Ex. 1077, 39:1–3 ("The main limitation [with post hoc analysis] is that this -- it is a less strict, less stringent analysis than the main prespecified analysis."), 42:13–15 ("a subgroup analysis – a non-prespecified or prespecified subgroup analysis is less strong than the main analysis.").

For the reason discussed above, we determine that Patent Owner's evidence of unexpected results are weak and entitled to little weight.

vii. Claim 1 Summary

Based upon our review of the parties' arguments and supporting evidence in the full record, we determine that Petitioner by a preponderance of the evidence has demonstrated that claim 1 is unpatentable as obvious over the combination of Nidorf, the COLCOT Protocol, the ESC Guidelines, and COLCRYS.

c. *Claim 8*

Petitioner contends that claim 8 is unpatentable as obvious over Nidorf, the COLCOT protocol, the ESC guidelines, and COLCRYS. Pet. 34–44. Patent Owner opposes. PO Resp. 32–48.

i. (Preamble 8) A method of treating a patient a[f]ter having a myocardial infarction (MI),

Based upon our review of the parties' arguments and supporting evidence in the full record in conjunction with the reasons discussed above (*see* § II.D.b.i), we determine that Petitioner by a preponderance of the evidence has demonstrated that the combination of Nidorf, the COLCOT protocol, and the ESC guidelines teach "[a] method of treating a patient a[f]ter having a myocardial infarction (MI)" as recited in preamble 8.

ii. (Element 8a) the method comprising initiating the administration of colchicine at a daily low dose to the patient

Based upon our review of the parties' arguments and supporting evidence in the full record in conjunction with the reasons discussed above (*see* § II.D.b.ii), we determine that Petitioner by a preponderance of the evidence has demonstrated that the combination of Nidorf and the COLCOT protocol teach a "method comprising initiating the administration of colchicine at a daily low dose to the patient" as recited in element 8a.

iii. (Element 8b) within about 3 days of the MI

Based upon our review of the parties' arguments and supporting evidence in in the full record in conjunction with the reasons discussed above (*see* § II.D.b.iv), we determine that Petitioner by a preponderance of the evidence has demonstrated that the combination of Nidorf and the COLCOT protocol teach the administration of colchicine within 0–30 days of the myocardial infarction, thereby overlapping with the administration between 0–3 day of the myocardial infarction as recited in element 8b.

iv. (Element 8c) wherein the colchicine is administered without pre-loading the patient with a higher dose of colchicine before the administration at the daily low dose

Based upon our review of the parties' arguments and supporting evidence in the full record in conjunction with the reasons discussed above (*see* § II.D.b.iii), we determine that Petitioner by a preponderance of the evidence has demonstrated that the combination of Nidorf and the COLCOT protocol teach the administration of colchicine without a pre-loading dose as recited in element 8c.

v. (Element 8d) wherein the patient is at a lower risk of a cardiovascular event, relative to a patient not being administered colchicine

For the reason discussed above (*see* § II.A.3), we find that the clause "wherein the patient is at a lower risk of a cardiovascular event, relative to a patient not being administered colchicine" of element 8d is not limiting because it simply expresses the intended result of the positively recited method step within the body of the claim. The method steps are preformed the same way whether or not the patient experiences a lower risk of a cardiovascular event. *See, e.g., In re Hirao*, 535 F.2d at 70 ("the preamble merely recites the purpose of the process; the remainder of the claim . . . does not depend on the preamble for completeness, and the process steps are able to stand alone.")

We have reviewed the parties' contentions and agree with Petitioner that this wherein clause merely describes the benefit of the administration of colchicine to the identified patient population. Because this clause merely defines the result of the positively recited method step of administering colchicine to patients after suffering MI we agree with Petitioner that element 8d is met by the combination of references. *See* Pet. 23; *see, e.g.*, *Minton*, 336 F.3d at 1381 (a "whereby clause in a method claim is not given weight when it simply expresses the intended result of a process step positively recited.").

Based upon our review of the parties' arguments and supporting evidence in the full record, we determine that Petitioner demonstrated by a preponderance of the evidence that the combination of Nidorf and the COLCOT protocol teach the recited purpose that "the patient is at a lower risk of a cardiovascular event, relative to a patient not being administered colchicine" as recited in element 8d.

vi. (Element 8e) wherein the cardiovascular event is resuscitated cardiac arrest, stroke, or urgent hospitalization for angina requiring coronary revascularization.

Petitioner contends that Nidorf teaches this limitation. Pet. 37–38 (citing Ex. 1007, 18:31–34, 18:65–67, 19: 6–14, 19:55–59, 19:63–64; Ex. 1049 ¶¶ 28, 39, 68, 69, 143).

Patent Owner does not specifically address this limitation. *See* PO Resp.

Nidorf teaches that "[t]his trial demonstrates that the addition of colchicine 0.5 mg/day to standard therapy in patients with stable coronary disease significantly reduces the risk of a cardiovascular event, including an ACS, out of hospital cardiac arrest and non-cardio-embolic ischemic stroke." Ex. 1007, 21:55–59, *see id.* at 22:12–24 (claims 1–3).

> An acute coronary syndrome (ACS) was defined as either (a) Acute Myocardial Infarction (AMI), as evidenced by acute ischemic chest pain associated with a rise in serum troponin above the upper limit of normal or (b) Unstable Angina (UA), as evidenced by a recent acceleration of the patient's angina unassociated with a rise in serum troponin but associated with angiographic evidence of a change in the patient's coronary anatomy.

Ex. 1007, 19:6–14.

The COLCOT protocol teaches administration of 0.5 mg colchicine per day to patients that suffered an MI "and after having completed any planned percutaneous revascularization procedures associated with their initial infarction." Ex. 1006, 1.

Based upon our review of the parties' arguments and supporting evidence in the full record, we determine that Petitioner by a preponderance of the evidence demonstrated that at least the combination of Nidorf and COLCOT protocol teaches treating patients with the identified cardiovascular event as encompassed by element 8e.

vii. Objective Indicia of Nonobviousness

For the reasons discussed above (II.D.4.b.vi), we determine that Patent Owner's evidence of unexpected results are weak and entitled to little weight.

viii. Summary

Based upon our review of the parties' arguments and supporting evidence in the full record, we determine that Petitioner by a preponderance of the evidence has demonstrated that claim 8 is unpatentable as obvious over Nidorf, the COLCOT protocol, the ESC guidelines, and COLCRYS.

d. *Dependent Claims 2–7, 9–26, and 28–50*

Claims 2–7, 9–26, and 28–50 depend directly or indirectly from claims 1, 8, or 27. Ex. 1001, 41:37–49, 41:61–42:32, 42:37–43:20. Petitioner provides citations in Nidorf, the COLCOT protocol, the ESC guidelines, and COLCRYS where each of dependent claims 2–7, 9–26, and 28–50 limitations are either disclosed or rendered obvious by the references. Pet. 46–54. Petitioner supports its assertions with the testimony of Dr. Ridker. *Id.* (citing Ex. 1049, ¶¶ 181–216).

Patent Owner disagrees. PO Resp. 51–55. We address Patent Owner's contentions below.

i. Claims 2–5, 29–32, 40–43 (Additional Medication)

Claims 2–5, 29–32, 40–43 depend either directly or indirectly from independent claims 1, 8, or 27 and further recite: "wherein the patient was prescribed a medication," "wherein the medication is an antiplatelet agent," "wherein the medication is aspirin," or "wherein the medication is a statin." Ex. 1001, 41:35–44, 42:40–47, 42:64–43:41.

Patent Owner contends that the claims are not obvious because adding colchicine to standard secondary prevention unexpectedly prevented future CV event. PO Resp. 53. For the reason discussed above with we are not persuaded by Patent Owners unexpected results argument. *See* II.D.4.d.

The COLCOT protocol recites that a "[p]atient must be treated according to national guidelines (including anti-platelet therapy, statin, renin-angiotensin-aldosterone system inhibitor (preferably angiotensin-converting enzyme) and beta-blocker when indicated)." Ex. 1006, 4. Petitioner contends that "a POSA would have immediately envisaged aspirin—the most common cardiovascular antiplatelet therapy as of 2019—from COLCOT Protocol's disclosure of "antiplatelet therapy" according to "national guidelines." Pet. 32–33 (citing Ex. 1049, ¶ 135; *see*, e.g., Ex. 1021, 3 ("A dose of 160 to 325 mg [aspirin] should be given on day 1 of AMI and continued indefinitely on a daily basis"); Ex. 1009, 31 ("For long-term prevention, low aspirin doses (75–100mg) are indicated")), *id.* at 48–49 (citing Ex. 1007, 20:6–10 ("therapy with aspirin, clopidogrel or both, high dose statin therapy (defined as a dose of statin equivalent to atorvastatin of 40 mg or more), beta blockers, calcium blockers and ACE inhibitors"), 21:25–54; Ex. 1020, 13, Table 1; Ex. 1049 ¶¶ 187–188).

Based on these disclosures in Nidorf and the COLCOT protocol and for the reasons stated in the Petition, which we adopt as our own findings, we are persuaded that Petitioner has demonstrated by a preponderance of evidence that claims 2–5, 29–32, and 40–43 are obvious. *See* Pet. 32–33, 48–49.

ii. Claims 6, 7, 9–11, 50 (Cardiovascular Event)

Claims 6, 7, 9–11, and 50 depend either directly or indirectly from claims 1, 8, or 27 and further recite: "wherein the patient is at a lower risk of a cardiovascular event, relative to a patient not being administered colchicine," "wherein the cardiovascular event is an ischemic cardiovascular event," "wherein the cardiovascular event is myocardial infarction," "wherein the cardiovascular event is stroke," "wherein the cardiovascular event is urgent hospitalization for angina requiring coronary revascularization," and "wherein the cardiovascular event is resuscitated cardiac arrest." Ex. 1001, 41:45–49, 41:61–67, 43:19–20.

Patent Owner makes the same arguments presented for claims 1 and 8. Specifically, that Nidorf's teaching for reduction of CV events with the treatment regimen disclosed in the LoDoCo trial is applicable only to patients that are stable for 6 months before initiating treatment. PO Resp. 50–51. For the same reason discussed above with respect to claims 1 and 8 we are not persuaded by Patent Owners argument. *See* II.D.4.b–c.

Nidorf teaches that

> [f]urther analysis confirmed that patients randomized to treatment were less likely to present with an [acute coronary syndrome] ACS unrelated to stent disease.
>
> The reduction in the primary outcome was largely driven by the reduction in the number of patients presenting with an ACS, (13/282 (4.6%) vs. 34/250 (13.4%), hazard ratio 0.33; 95% CI; 0.18-0.63; p<0.001). Out of hospital cardiac arrest and non-

> cardio-embolic ischemic stroke were infrequent but were also reduced in the treatment group.

Id. 20:39–44, 48–50.

The COLCOT protocol instructs to initiate colchicine treatment within 30 days of an MI event (i.e. between 0–30). *Id.* at 1. We credit the testimony of Dr. Ridker that based on the ESC guidelines and what was known to a person of ordinary skill of art at the time the invention was made about the inflammatory involvement in coronary artery disease that "initiating treatment earlier in time would have provided a reasonable expectation of success." Pet. 48 (citing Ex. 1049 ¶¶ 168 ("CANTOS provided 'definitive evidence that directly targeting inflammation is beneficial for the secondary prevention of cardiovascular disease.'"), 169; Ex. 1049 ¶ 184.

We are persuaded, for the reasons stated in the Petition, which we adopt as our own findings, that Petitioner has demonstrated by a preponderance of evidence that claims 6, 7, 9–11, and 50 are obvious. *See* Pet. 47–48.

iii. Claims 12, 26, 33, 39, 44, 49 (Patient Population)

Claims 12, 26, 33, 44, and 49 depend either directly or indirectly from claims 1, 8, or 27 and further recite: "wherein the patient has atherosclerotic coronary artery disease" and "wherein the patient is an adult human." Ex. 1001, 42:1–2, 42:31–32, 42:48–49, 43:5–6, 43:17–18.

Patent Owner contends that these claims are not obvious for the same reason that the independent claims are not obvious. PO Resp. 53.

The COLCOT protocol recites that "[a]therosclerosis is the most common cause of myocardial infarction, stroke and peripheral arterial disease." Ex. 1006, 4. The COLCOT protocol included "[m]ales and females of at least 18 years of age capable and willing to provide informed consent." *Id.*

Nidorf teach that "[d]espite routine use of anti-platelet and statin therapy, patients with atherosclerotic vascular disease continue to be at risk of cardiovascular events, possibly because these treatments fail to target some of the inflammatory pathways implicated in the disease." Ex. 1007, 2:34–38. Nidorf acknowledges that "many of these [known] treatments are recommended for acute conditions and do not address or provide for a long-term reduction in cardiovascular events in patients with clinically stable atherosclerotic vascular disease." *Id.* at 2:45–48.

Petitioner contends that the recitation of "the patient has atherosclerotic coronary artery disease" merely identifies an underlying disease known to be common in MI patients. Pet. 50. We credit Dr. Ridker's testimony "that atherosclerosis is the most common cause of myocardial infarction and that the vast majority of MI patients would have atherosclerotic coronary artery disease." Ex. 1049 ¶ 194 (citing Ex. 1005, 1; Ex. 1006, 1); Ex. 1015, 1 ("Atherosclerosis is the central disease process underlying most instances of myocardial infarction (MI), ischemic stroke, and peripheral artery disease").

Based on these disclosures in Nidorf and the COLCOT protocol and for the reasons stated in the Petition, which we adopt as our own findings, we are persuaded that Petitioner has demonstrated by a preponderance of evidence that claims 12, 26, 33, 39, 44, and 49 are obvious. *See* Pet. 50.

iv. Claims 13, 36, 46 (Initiation Upon Assessment)

Claims 15, 36, and 46 depend either directly or indirectly from claims 1, 8, or 27 and further recite "wherein the administration of colchicine is initiated upon assessment in (a) an emergency department (ED), (b) the hospital, or (c) a medical office setting." Ex. 1001, 42:3–6.

Patent Owner contends that these claims are not obvious for the same reason that the independent claims are not obvious. PO Resp. 53–54.

Petitioner contends that this limitation "covers initiating colchicine treatment prior to patient discharge from the typically about 3-day post-MI hospitalization." Pet. 51 (citing Ex. 1049 ¶¶ 84, 118, 158, 159, 200–204). Petitioner contends that the "ESC Guidelines further recommended initiating secondary prevention treatments during a hospital stay, which would necessarily require assessment of the patients in the hospital before treatment initiation." Pet. 51. Patients with MI frequently show up in a hospital emergency room. For the reasons discussed above (*see* II.4.b.iv), we agree with Petitioner that patient compliance with taking medicine is a reason for initiating treatment before they are discharged from a hospital. *See* Pet. 42 (citing Ex. 1009, 33 ("Lipid-lowering treatment should be started as early as possible, as this increases patient adherence after discharge, and given as high-intensity treatment, as this is associated with early and sustained clinical benefits"); Ex. 1049 ¶ 171).

We are persuaded, for the reasons stated in the Petition, which we adopt as our own findings, that Petitioner has demonstrated by a preponderance of evidence that claims 13, 36, and 46 are obvious. *See* Pet. 51.

v. Claims 17–25, 34–35, 45 (Dose and Frequency)

Claims 17–25, 34–35, and 45 depend either directly or indirectly from claims 1, 8, or 27 and further recite: "wherein the colchicine is administered at 0.3 to 0.7 mg," "wherein the colchicine is administered at 0.4 to 0.6 mg," "wherein the colchicine is administered at 0.5 mg," "wherein the colchicine is administered once, twice or three times a day," and "wherein the co[l]chicine is administered once per day." Ex. 1001, 42:12–30, 42:50–54, 54–57, 43:7–8.

Patent Owner contends that these claims are not obvious for the same reason that the independent claims are not obvious. PO Resp. 54.

We are not persuaded by Patent Owner's contentions. In the COLCOT protocol patients who suffered an acute MI within the last 30 days received either 0.5 mg/day colchicine or a matching placebo for about 2 years or until "the target of 301 primary endpoints has been reached." Ex. 1006, 1. Nidorf similarly teaches administering to patients with cardiovascular disease "a daily dose of colchicine of between about 0.1 mg and about 0.75 mg or between about 0.1 mg and about 0.5 mg." Ex. 1007, 15:66–16:1.

Based on these disclosures in the COLCOT protocol and Nidorf, we are persuaded, for the reasons stated in the Petition, which we adopt as our own findings, that Petitioner has demonstrated by a preponderance of evidence that claims 17–25, 34–35, and 45 are obvious. *See* Pet. 51.

vi. Claims 14–16, 37, 38, 47, 48 (Tablet)

Claims 14–16, 37, 38, 47, and 48 depend either directly or indirectly from claims 1, 8, or 27 and further recite: "wherein the colchicine is in the form of a tablet," "wherein the tablet is coated," and "wherein the tablet is filmcoated." Ex. 1001, 42:7–12, 42:58–61, 43:13–16.

Patent Owner contends that these claims are not obvious for the same reason that the independent claims are not obvious. PO Resp. 54.

We are not persuaded by Patent Owner's contentions. The COLCOT protocol describes that the drug colchicine is taken once a day as 0.5 mg tablet. Ex. 1006, 2. Nidorf teaches that the dosage form of the active agent (colchicine) can include tablets or capsules. *See* Ex. 1007, 4: 41–45. "Colchicine is a tricyclic alkaloid and has a molecular mass of 399.437. The active ingredient colchicine as well as its tablet formulation is listed in various national and intonational pharmacopeias such as the United States Pharmacopeia (USP)." *Id.* at 6:25–29. COLCRYS teaches a colchicine dosage form "0.6 mg tablets — purple capsule-shaped, film-coated with AR 374 debossed on one side and scored on the other side." Ex. 1016, 4.

We are persuaded, for the reasons stated in the Petition, which we adopt as our own findings, that Petitioner has demonstrated by a preponderance of evidence that claims 13, 36, and 46 are obvious. *See* Pet. 51.

e. *Summary*

We conclude that Petitioner's evidence of obviousness of the challenged claims over the combination of Nidorf, the COLCOT protocol, the ESC guidelines, and COLCRYS, when considered and weighed with Patent Owner's weak evidence of secondary considerations, establishes by a preponderance of the evidence that claims 1–50 of the '063 patent are unpatentable under 35 U.S.C. § 103(a).

III. CONCLUSION

For the reasons given, based on the arguments and evidence of record, Petitioner has met their burden to prove by a preponderance of the evidence that claims 1–50 of the '063 patent are unpatentable as obvious.

In summary

Claim(s)	35 U.S.C. §	Reference(s)/Basis	Claim(s) Shown Unpatentable	Claim(s) Not Shown Unpatentable
27, 29–36, 39	102	COLCOT protocol	27, 29–36, 39	
1–50	103	Nidorf, COLCOT Protocol, ESC Guidelines, COLCRYS	1–50	
1–13, 17–26, 28, 40–46, 49, 50	102	Bouabdallaoui[11]		
1–26, 28, 40–50	103	Bouabdallaoui, COLCRYS[12]		
Overall Outcome			1–50	

[11] Because of our determination that Petitioner establishes by a preponderance of the evidence that claims 1–50 are obvious over the combination of Nidorf, the COLCOT Protocol, the ESC Guidelines, and COLCRYS, we do not reach this alternate challenge to the claims. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner "is entitled to a final written decision addressing all of the claims it has challenged"); *see also Boston Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App'x 984, 990 (Fed. Cir. 2020) (nonprecedential) (stating that the "Board need not address issues that are not necessary to the resolution of the proceeding," such as "alternative arguments with respect to claims [the Board] found unpatentable on other grounds").

[12] See above footnote 7.

IV. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Petitioner has proved by a preponderance of the evidence that claims 1–50 are unpatentable; and

FURTHER ORDERED that, because this is a Final Written Decision, parties to the proceeding seeking judicial review of the decision must comply with the notice and service requirements of 37 C.F.R. § 90.2.

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